M Build Guide

General Guidance I

3D Printing

- Level the bed and clean the nozzle&bed before each major print batch.
- Use recommended layer heights (e.g. 0.2 mm for strength vs. 0.1 mm for detail).
- Choose infill (20–30 %) for structural parts, lower for purely cosmetic pieces.
- Orient parts as shown
- Only enable supports where absolutely necessary; use tree-supports or interface layers for best print quality and ease of support removal
- Apply a brim or raft for small footprints to prevent warping.
- Group prints by build-step to avoid mix-ups later.
- Batch # provided are just for reference, please manage printing order based on printer size, printing time, confidence of printing multiple big components at same time, etc.

General Guidance II

Imbedding Heat-set Inserts

- Set soldering iron to ~200–220 °C
  (adjust based on plastic—lower for PLA, higher for ABS & PETG).
- Use a flat-chisel tip that matches the insert's outer diameter.
- Plunge the heated insert straight down—avoid twisting or rocking.
- Stop when the flange is flush with the plastic surface.

General Guidance III

Imbedding Magnets

- Dry-fit first to verify orientation and depth.
- Check polarity before insertion
- Apply adhesive sparingly; avoid overflow onto visible surfaces.
- Adhesive Choices: Gel type Super Glue or 2-part Epoxy

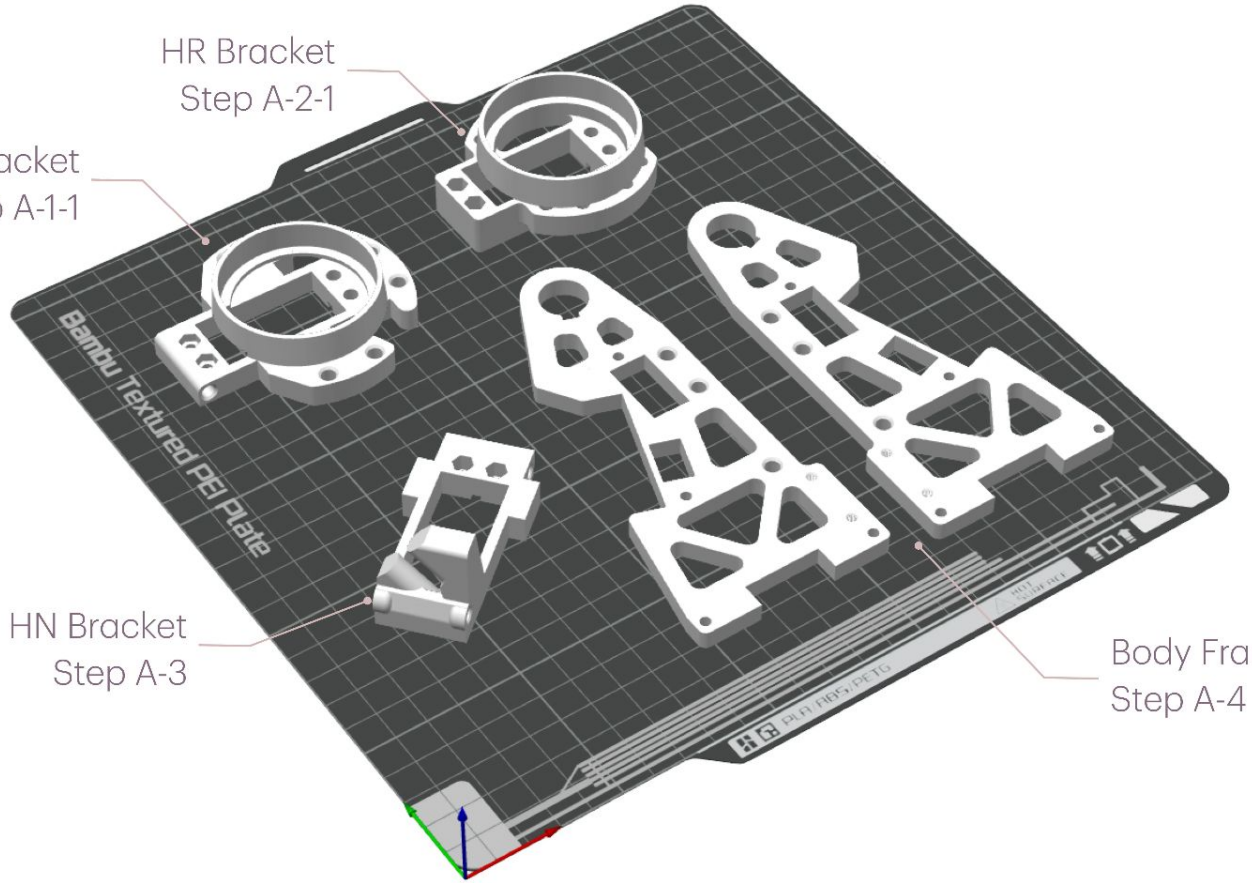
Part A: Module Pre-assembly

Embed Heat-set Insert & Magnets; Prepare Actuation Modules



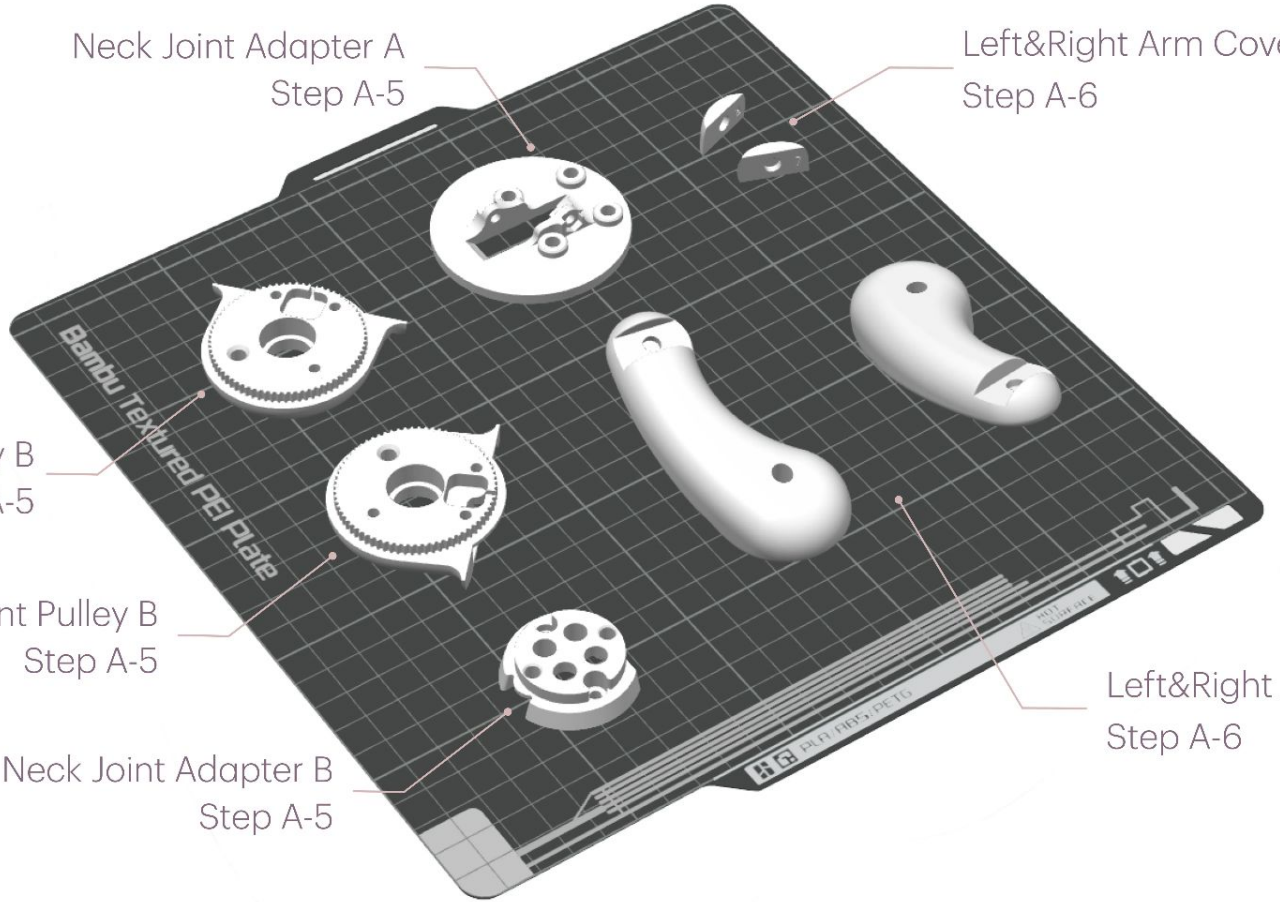
Step A-0-1: 3D Print Parts

Batch #1



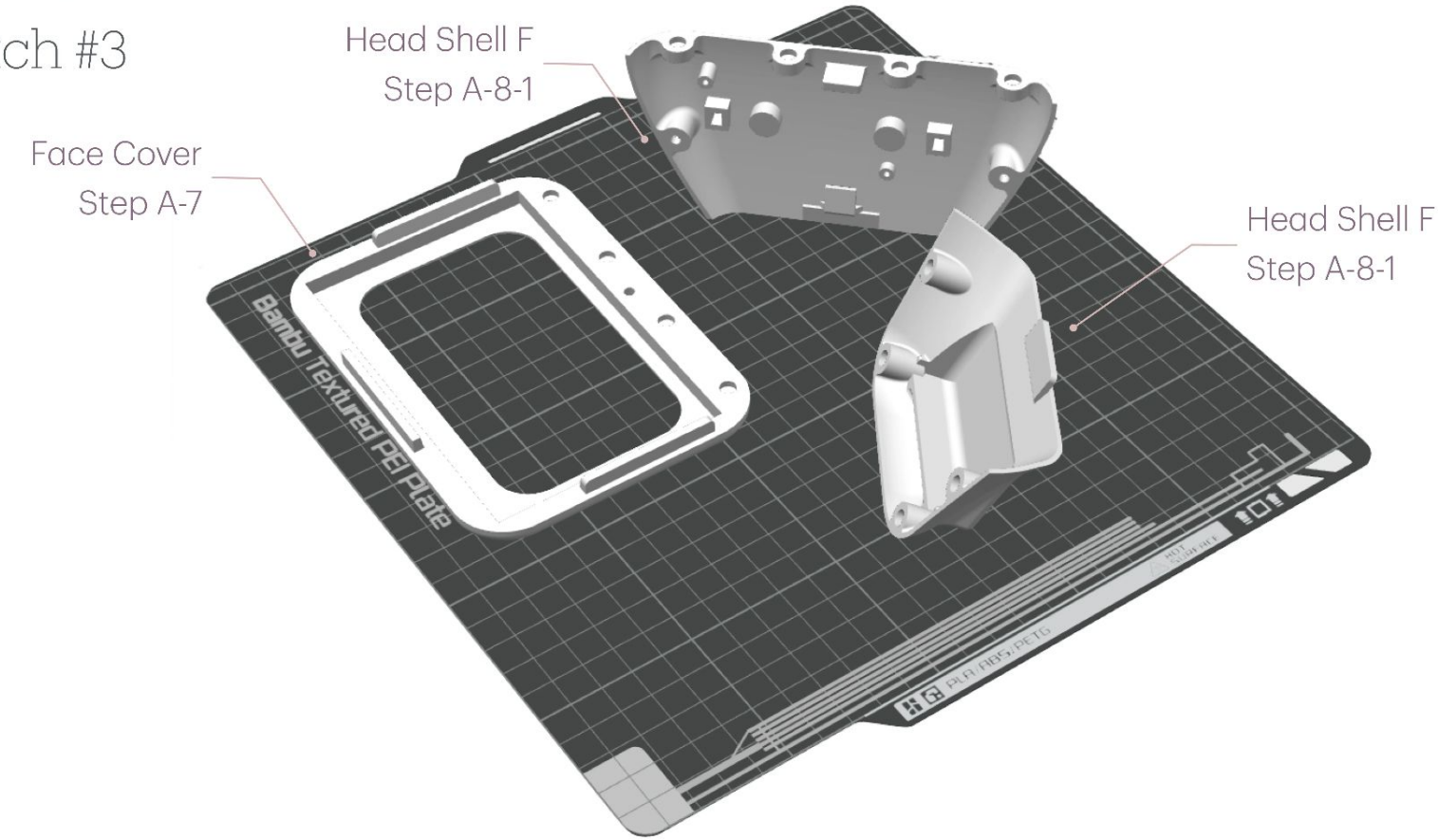
Step A-0-2: 3D Print Parts

Batch #2



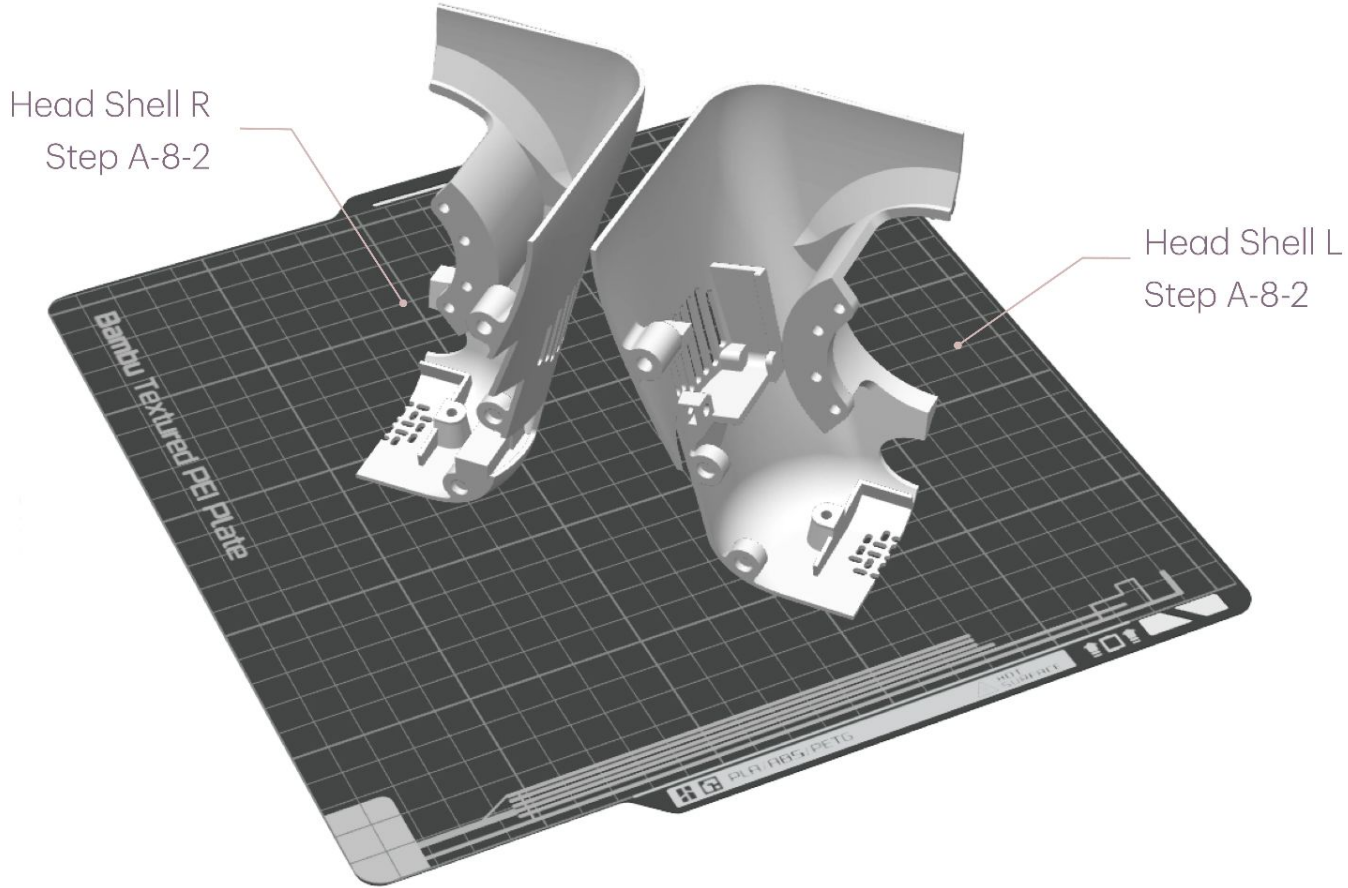
Step A-0-3: 3D Print Parts

Batch #3



Step A-0-4: 3D Print Parts

Batch #4



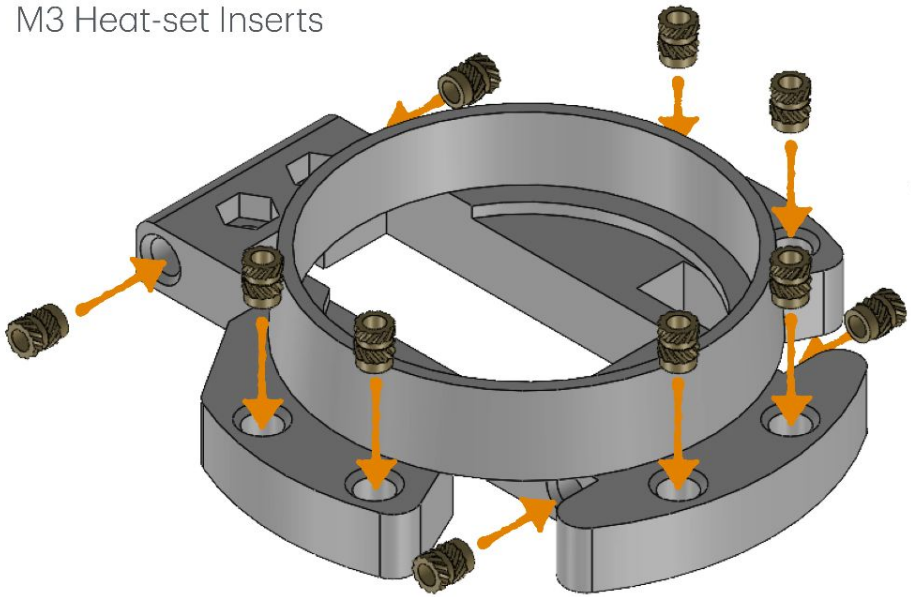
Step A-1-1: Base Rotation Module Assembly

Embed heat-set inserts

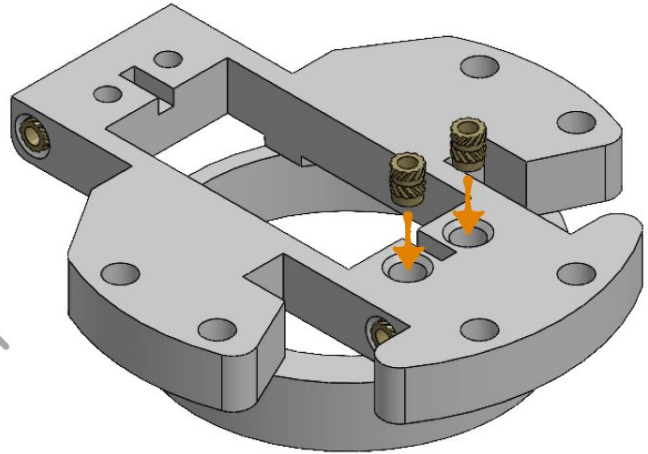
You need:

1x BR bracket

12x M3 Heat-set Inserts



Flip



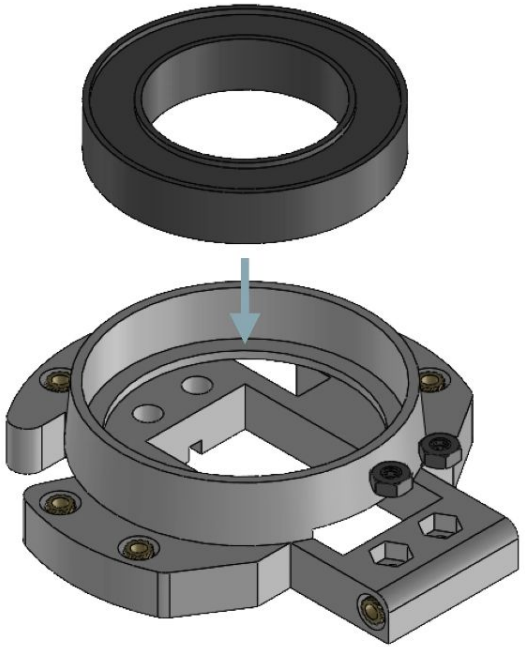
Heat-set

Step A-1-2: Base Rotation Module Assembly

Press bearing; Install servo motor

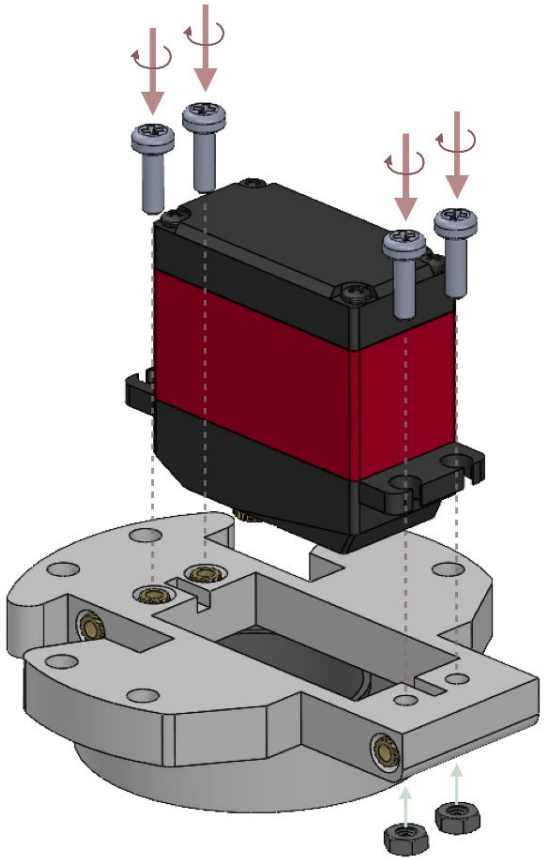
You need:

- 1x BR bracket (from Step A-1-1)
- 1x 6906 Bearing
- 1x Standard Servo
- 4x M3 * 12 Screw
- 2x M3 Nuts



Seat
Screw in
Press Fit

Flip



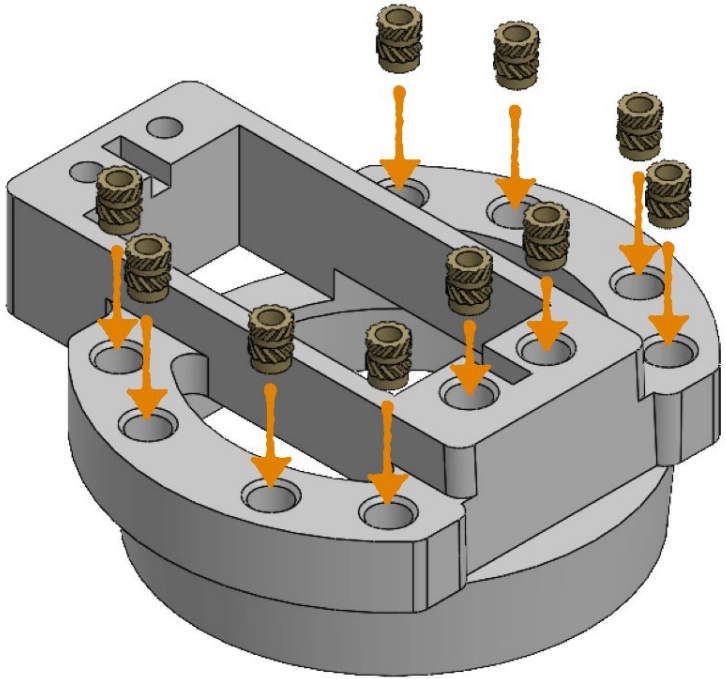
Step A-2-1: Head Rotation Module Assembly

Embed heat-set inserts

You need:

1x HR bracket

10x M3 Heat-set Inserts



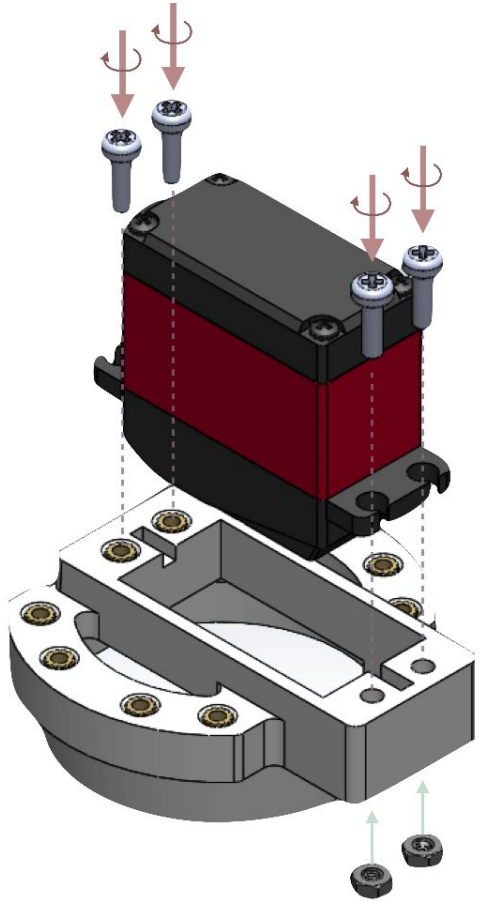
Heat-set

Step A-2-2: Head Rotation Module Assembly

Install servo motor

You need:

- 1x HR bracket (from Step A-2-1)
- 1x Standard Servo
- 4x M3 * 12 Screw
- 2x M3 Nuts



Seat

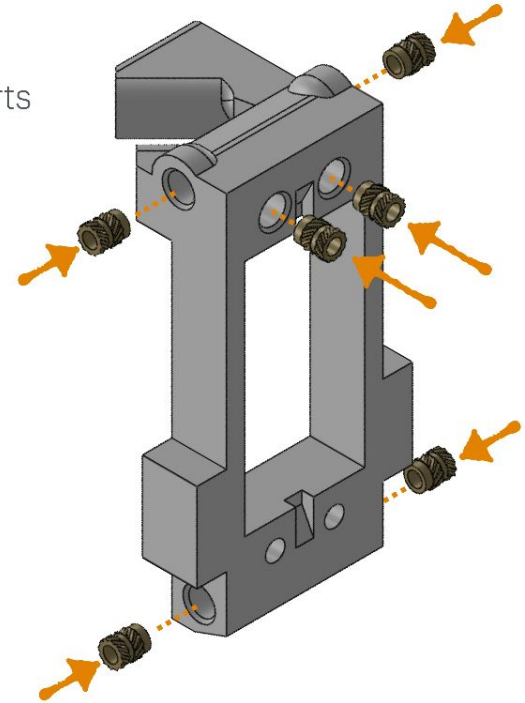
Screw in

Step A-3: Head Nod Module Assembly

Embed 6x heat-set inserts; Install servo motor

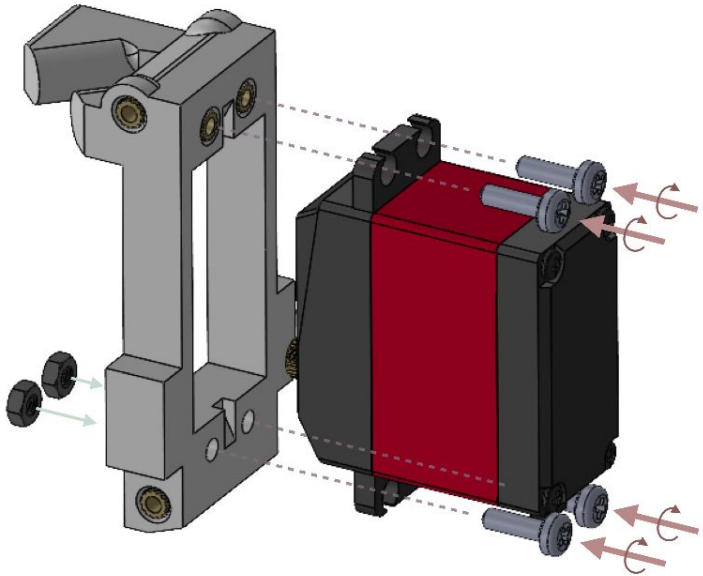
You need:

- 1x HN bracket
- 6x M3 Heat-set Inserts
- 1x Standard Servo
- 4x M3 * 12 Screw
- 2x M3 Nuts



Seat
Screw in
Heat-set

Rotate



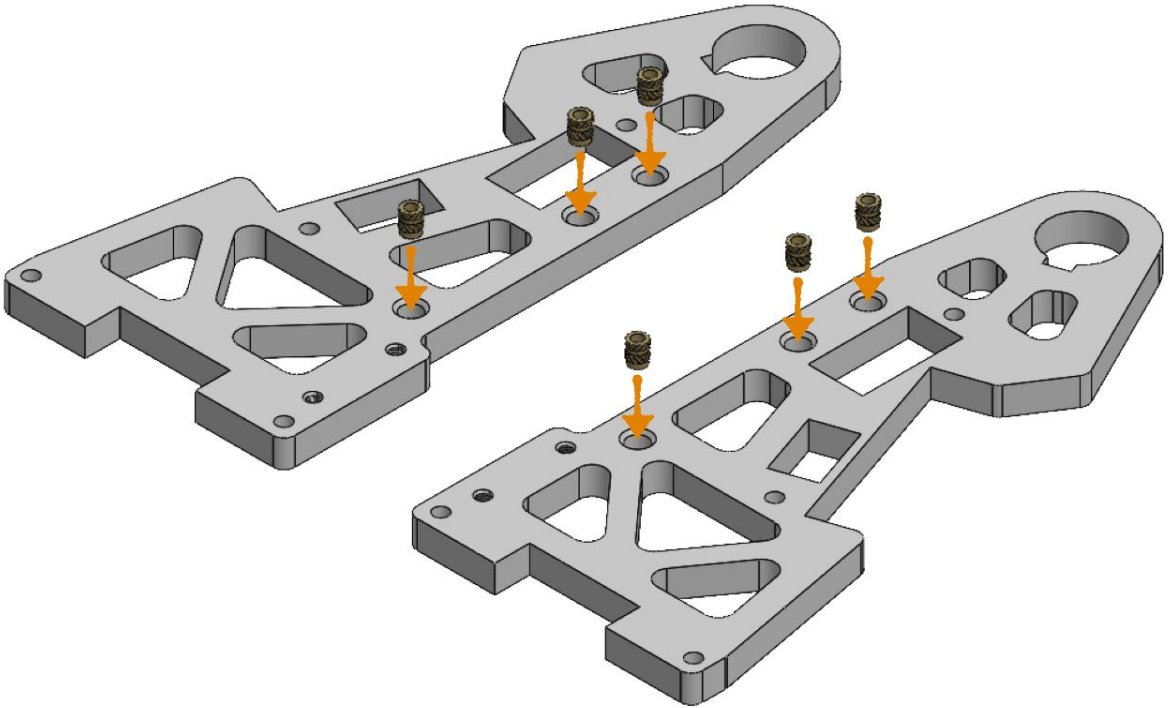
Step A-4: Prepare Body Frame

Embed heat-set inserts

You need:

2x Body Frame

6x M3 Heat-set Inserts



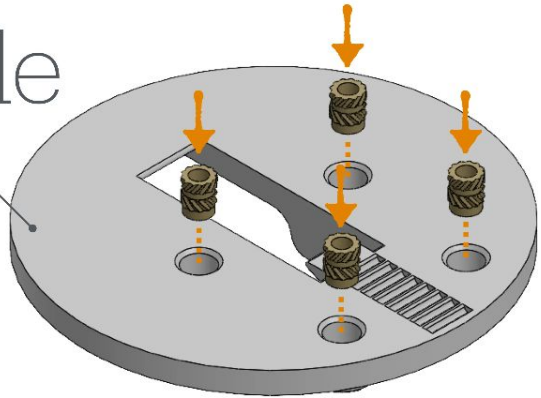
Heat-set

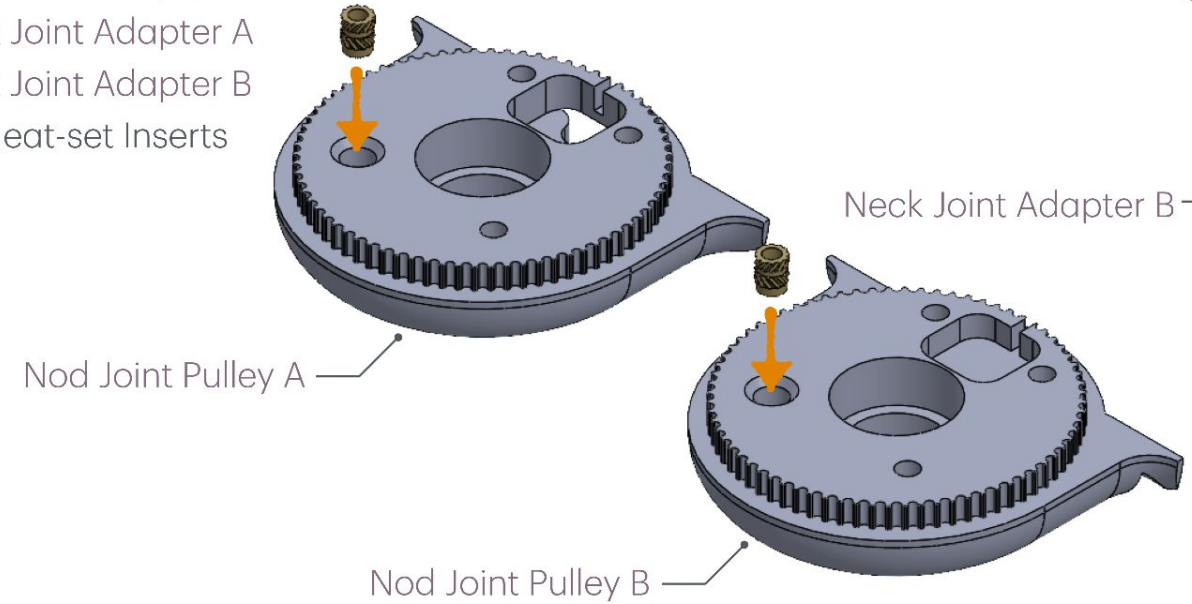
Step A-5: Prepare Neck Module

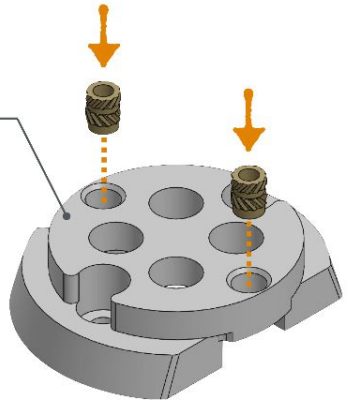
Embed heat-set inserts

You need:

- 1x Nod Joint Pulley A
- 1x Nod Joint Pulley B
- 1x Neck Joint Adapter A
- 1x Neck Joint Adapter B
- 6x M3 Heat-set Inserts







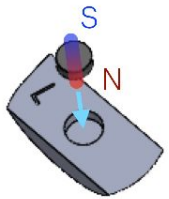
Heat-set →

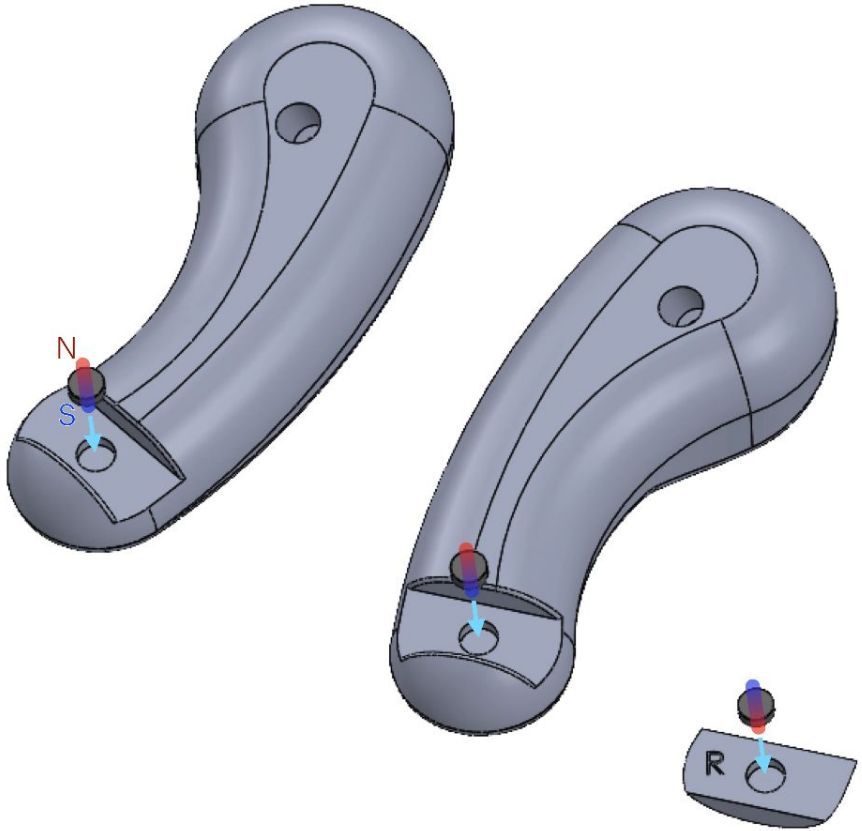
Step A-6: Prepare Arm Pieces

Embed magnets

You need:

- 1x Left Arm
- 1x Left Arm Cover
- 1x Right Arm
- 1x Right Arm Cover
- 4x D5*2mm Magnets





Glue →

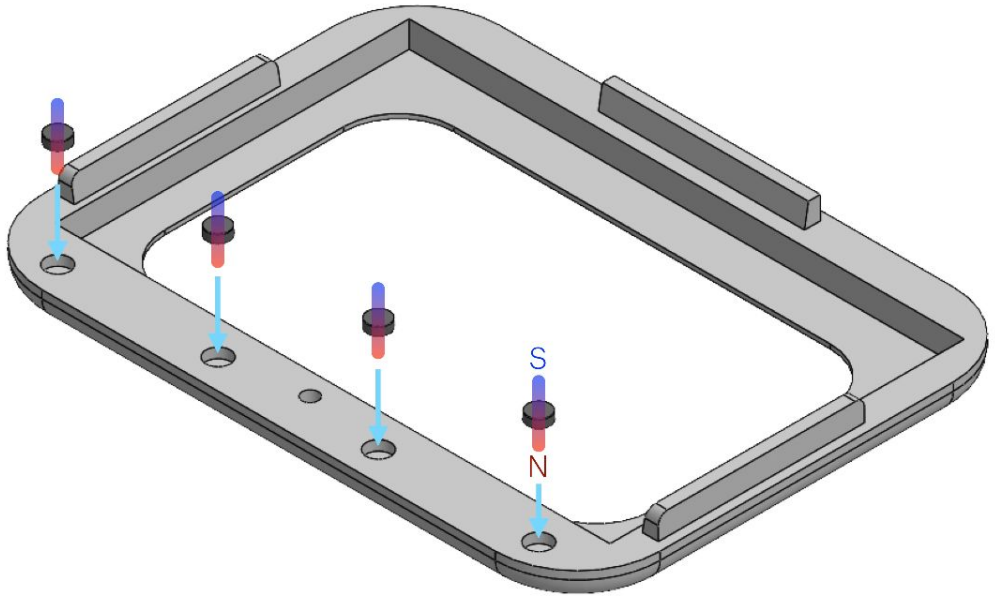
Step A-7: Prepare Face Cover

Embed heat-set inserts

You need:

1x Face Cover

4x D5*2mm Magnets



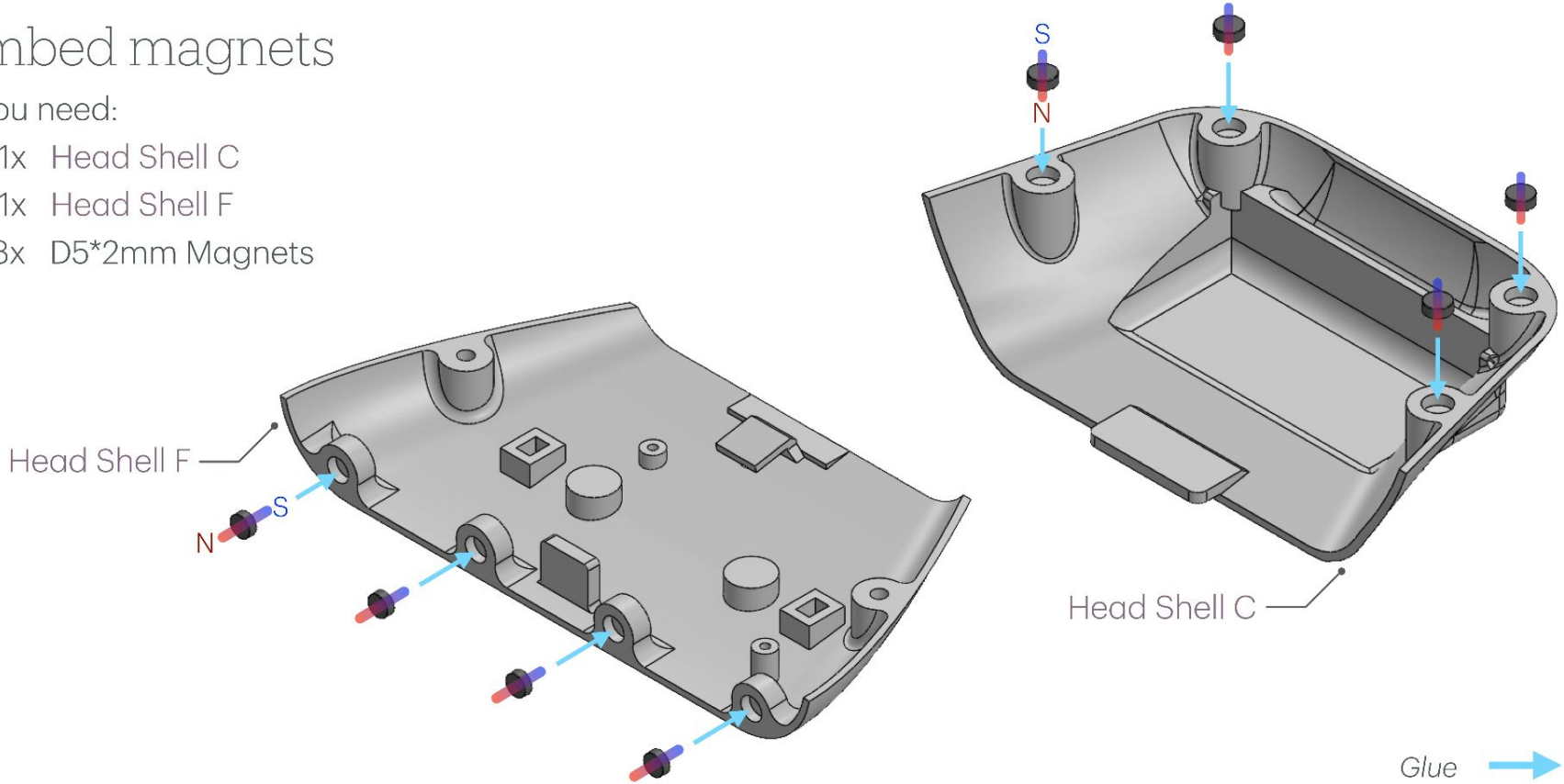
Glue

Step A-8-1: Prepare Head Shell Pieces

Embed magnets

You need:

1x Head Shell C
1x Head Shell F
8x D5*2mm Magnets

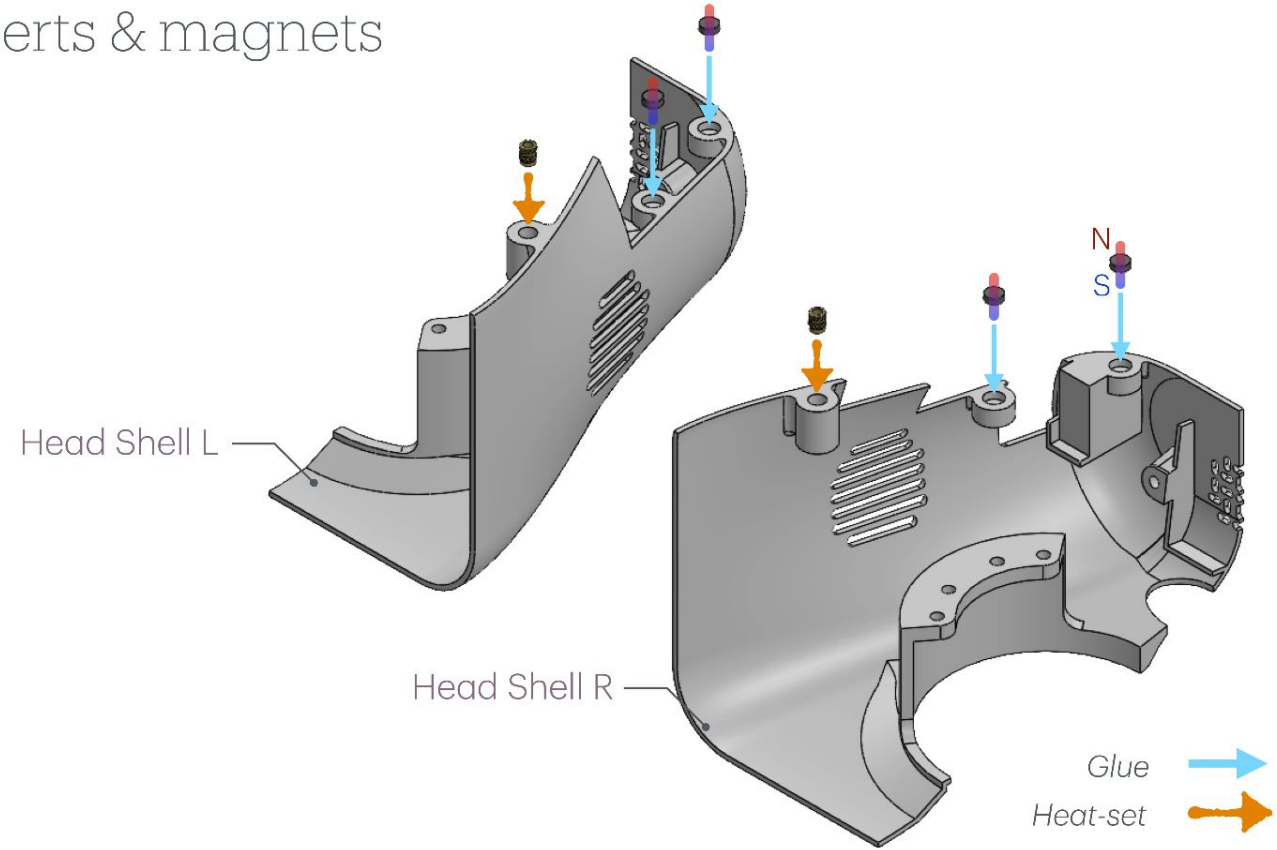


Step A-8-2: Prepare Head Shell Pieces

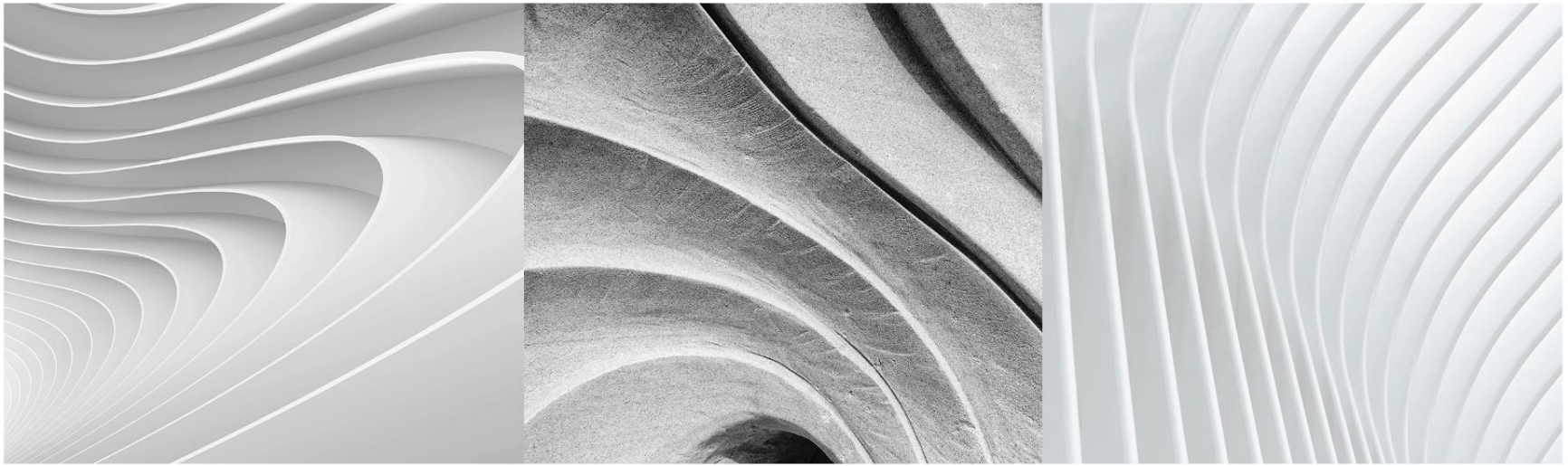
Embed heat-set inserts & magnets

You need:

- 1x Head Shell L
- 1x Head Shell R
- 2x M3 Heat-set Inserts
- 4x D5*2mm Magnets

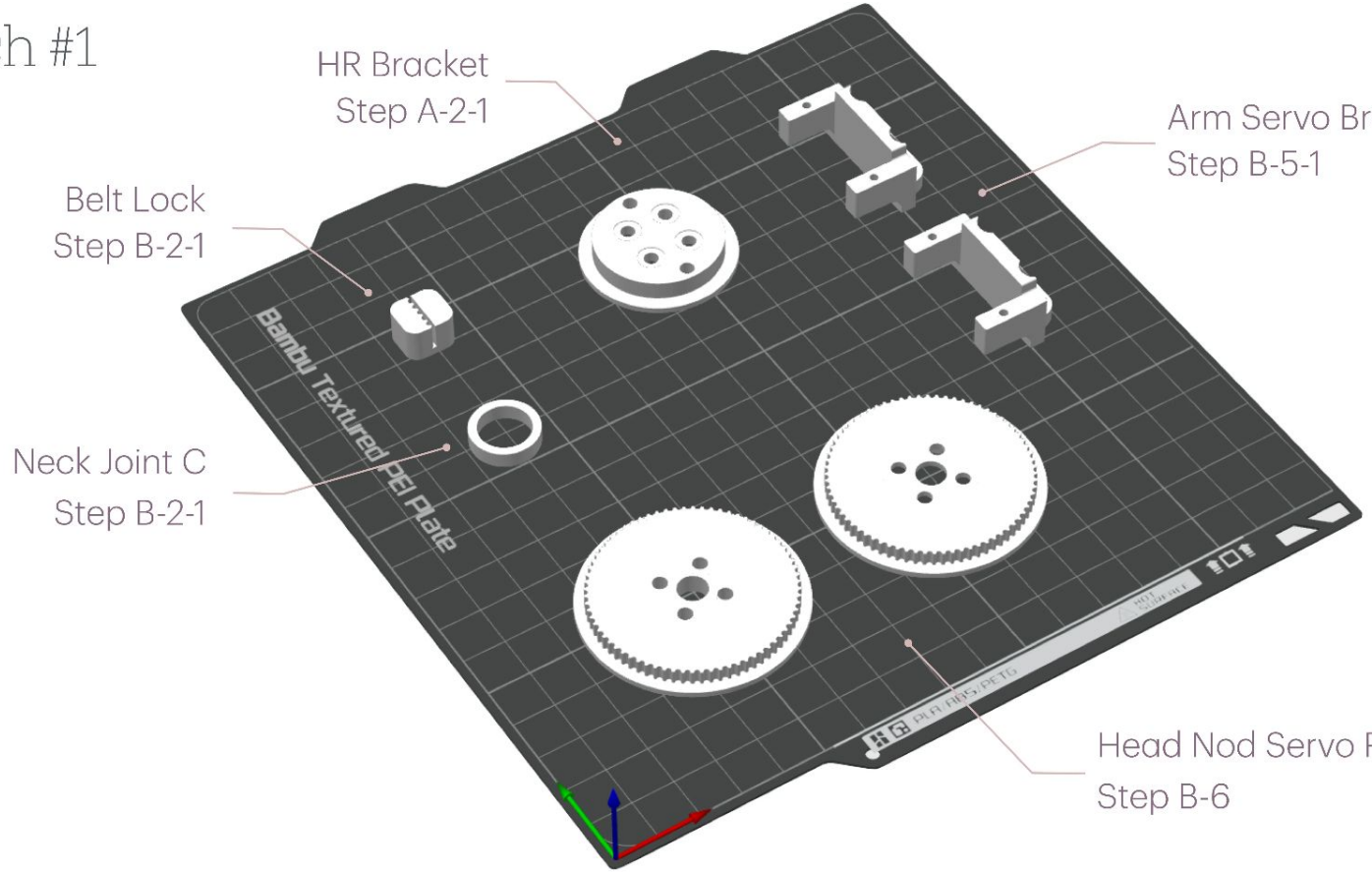


Part B: Body Skeleton Assembly



Step B-0: 3D Print Parts

Batch #1

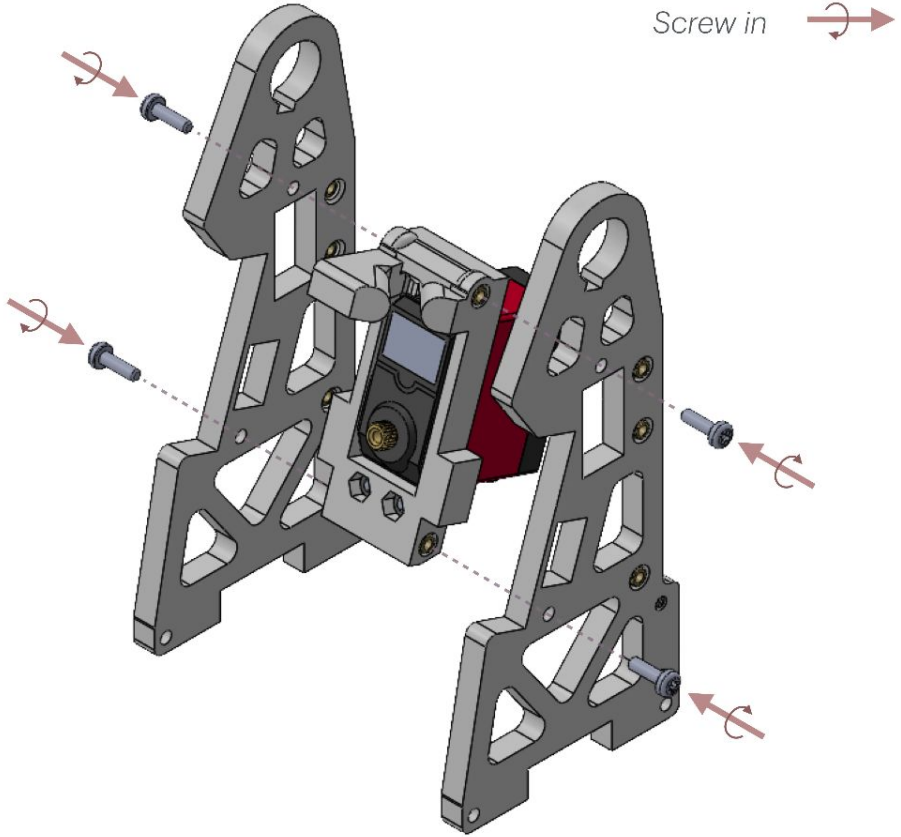


Step B-1-1: Assemble Body Frame Unit

You need:

2x Body Frame (from Step A-5)
1x HN Module (from Step A-4)
4x M3 * 12 Screw

Important: Wire management



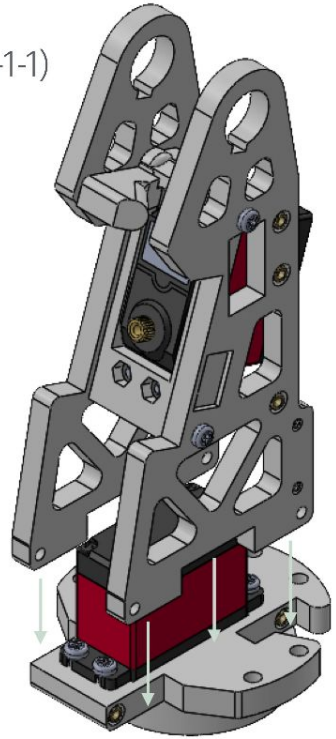
Step B-1-2: Assemble Body Frame Unit

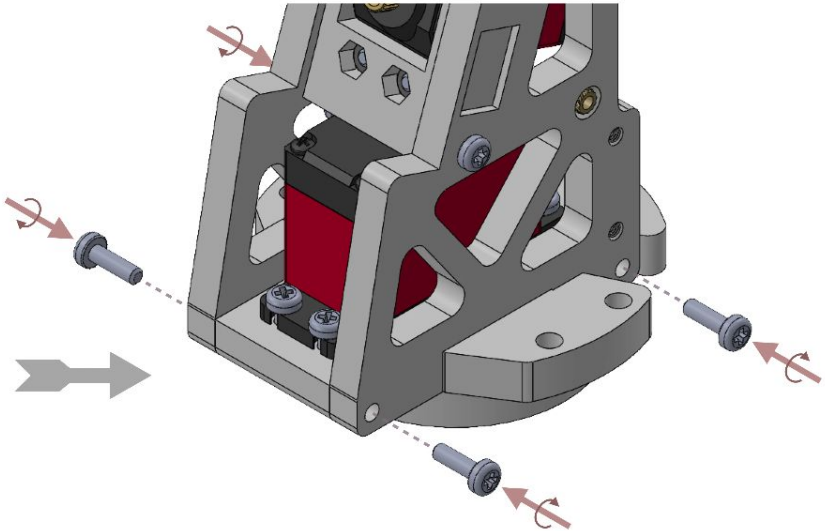
Screw in
Seat

You need:

1x Body Frame Unit (from Step B-1-1)
1x BR Module (from Step A-2-2)
4x M3 * 12 Screw

Important: Wire management





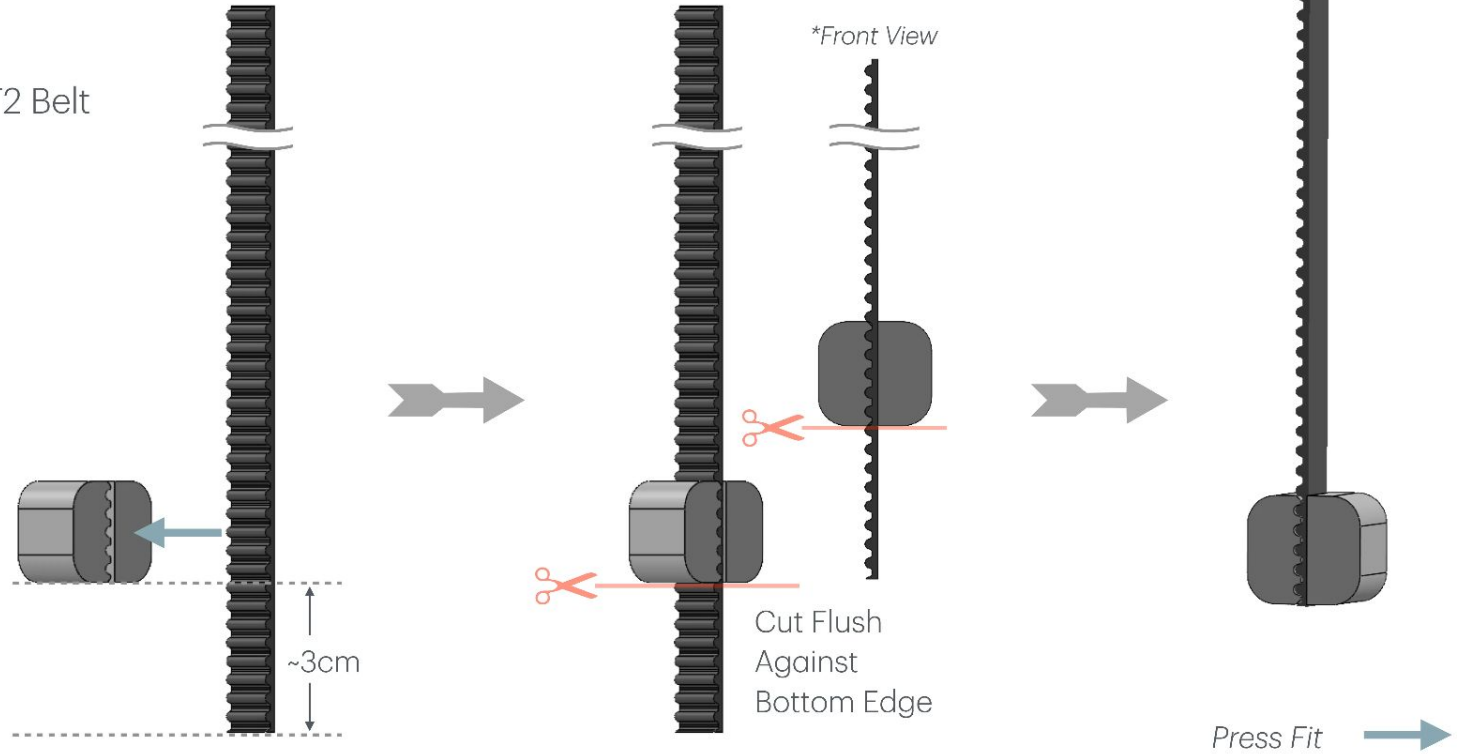
Step B-2-1: Prepare Belt

Important: Orient both part exactly as shown. Before you make the cut, double-check that the orientation matches correctly.

You need:

1x ~40cm GT2 Belt

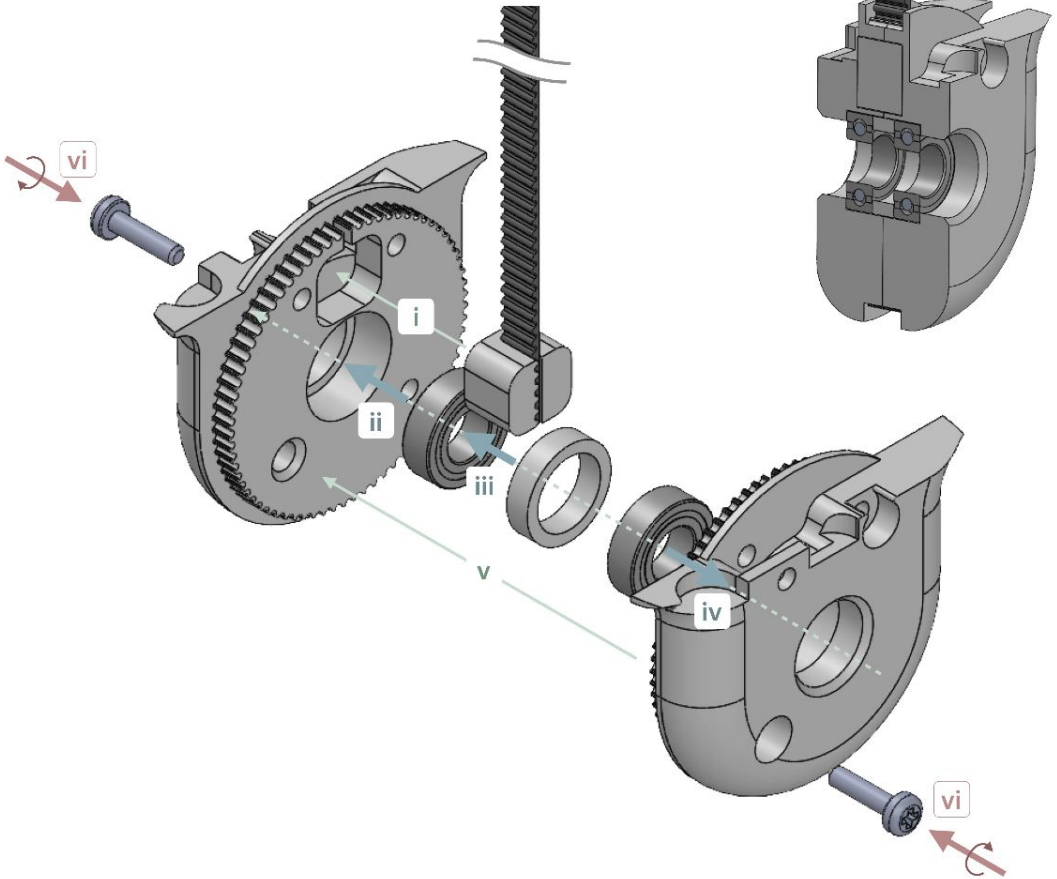
1x Belt Lock



Step B-2-2: Assemble Neck Joint

You need:

1x Neck Joint A (from Step A-5)
1x Neck Joint B (from Step A-5)
1x Neck Joint C
1x Assembled Belt (from Step A-9)
2x M3 * 12 Screw
2x 688 Bearing



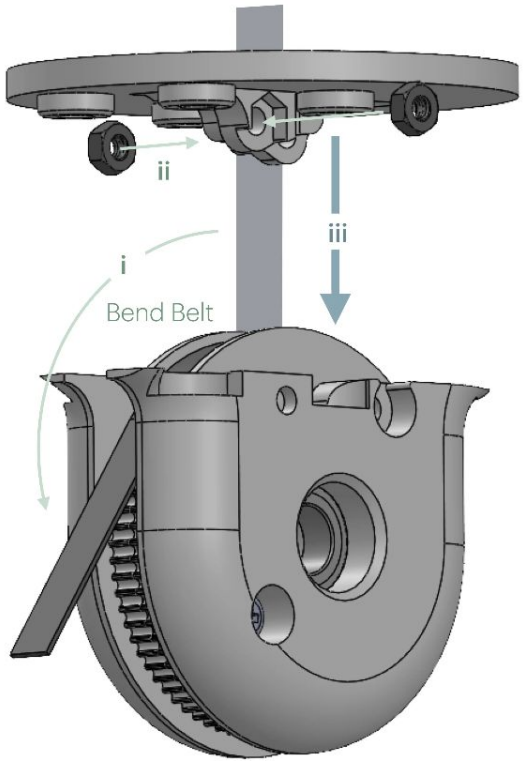
*Cross-section view after assembly

Seat
Screw in
Press Fit

Step B-2-3: Assemble Neck Joint

You need:

1x Nod Joint Pulley A (from Step A-5)
1x Neck Joint (from Step B-2-1)
2x M3 nut



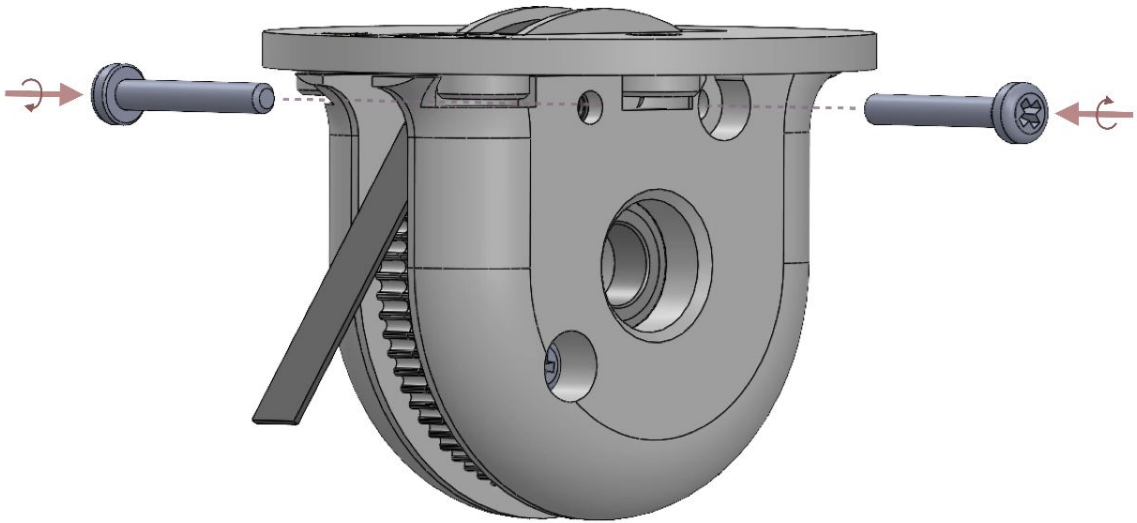
Seat
Screw in
Press Fit

Step B-2-4: Assemble Neck Joint

You need:

1x Neck Joint (from Step B-2-2)

2x M3 * 20 Screw

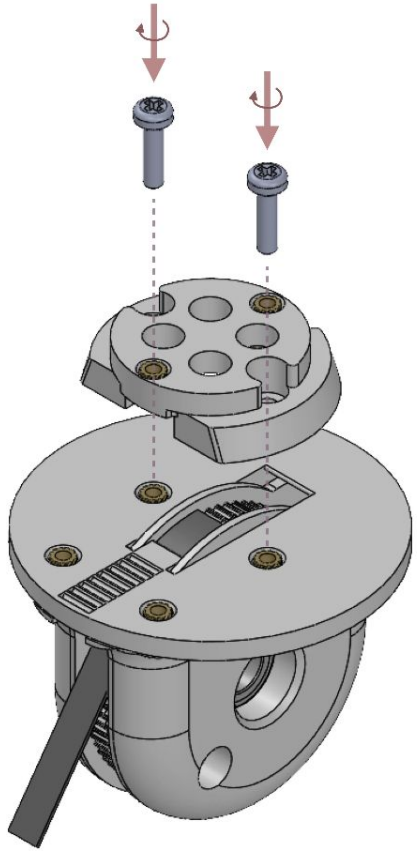


Screw in

Step B-2-4: Assemble Neck Joint

You need:

1x Nod Joint Adapter B (from Step A-5)
1x Neck Joint (from Step B-2-3)
2x M3 * 12 Screw

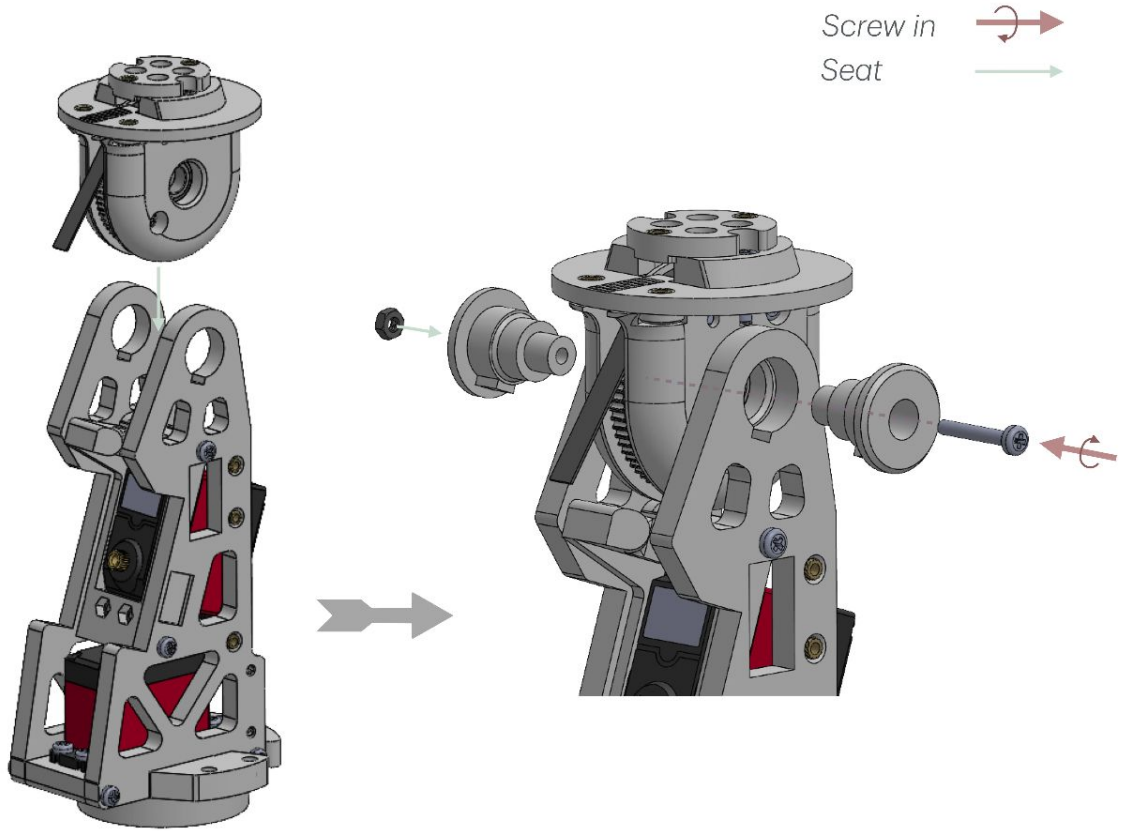


Seat
Screw in
Press Fit

Step B-3: Attach Neck Joint to Body

You need:

- 1x Body Frame Unit (from Step B-1-2)
- 1x Neck Joint (from Step B-2-4)
- 1x M3 * 20 Screw
- 1x M3 Nut

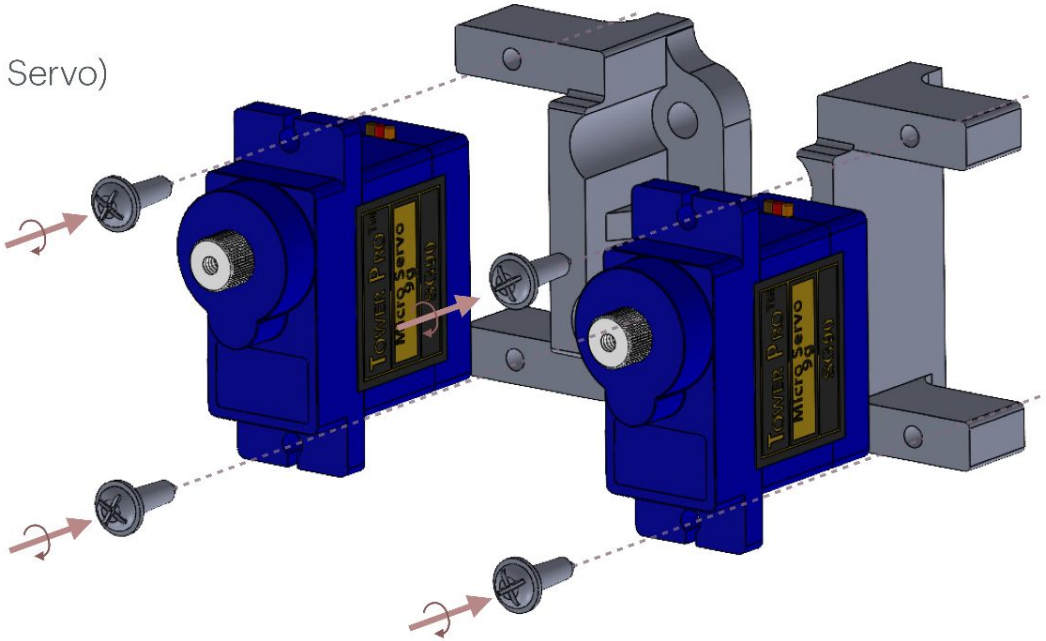


Step B-4-1: Arm Rotation Servo Brackets

Important: Orient the brackets and servos as shown; the outcome should be two mirrored assemblies

You need:

- 2x Arm Servo Bracket
- 2x Micro Servo
- 4x Micro Servo Screw (come with Servo)



Screw in

Step B-4-2: Arm Rotation Servo Brackets

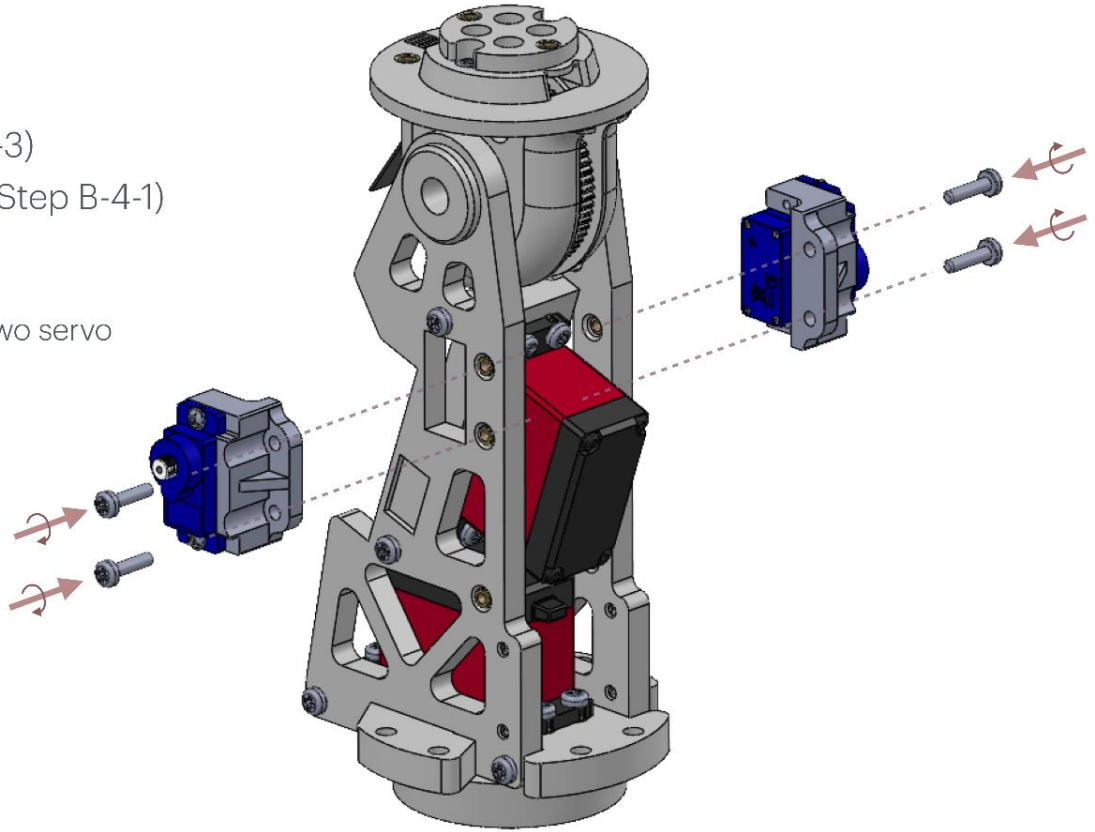
You need:

1x Body Frame Unit (from Step B-3)

2x Arm Rotation Assembly (from Step B-4-1)

4x M3 * 12 Screw

Important: Orient the assembly as shown; two servo rotation shaft should be aligned

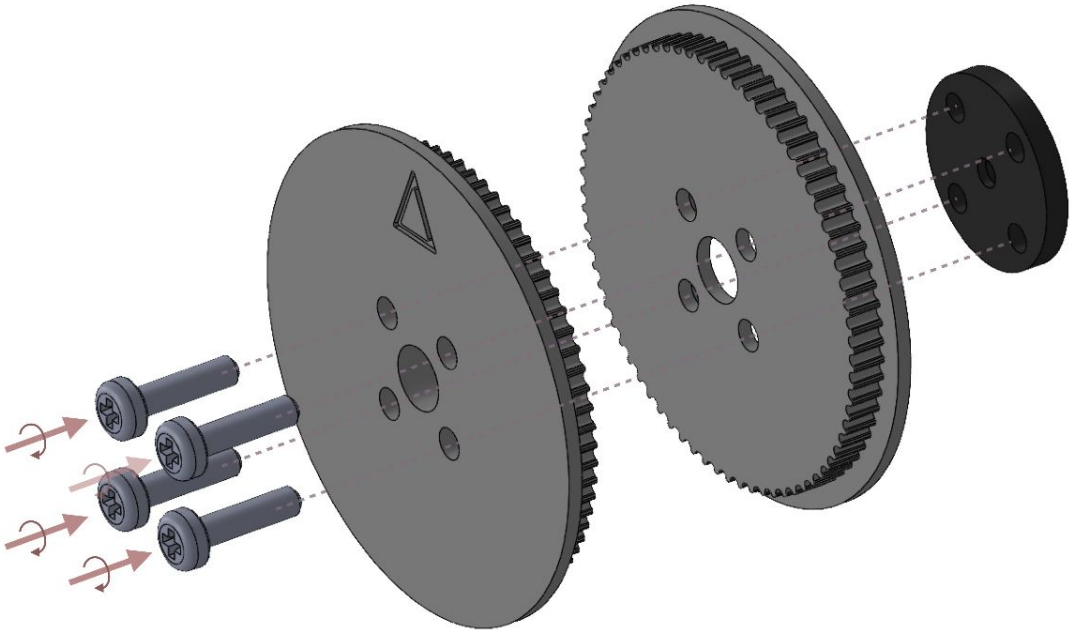


Screw in

Step B-5: Prepare Head Nod Pulley

You need:

- 1x Head Nod Servo Pulley A
- 1x Head Nod Servo Pulley B
- 1x Servo Disk
- 4x M3 * 12 Screw

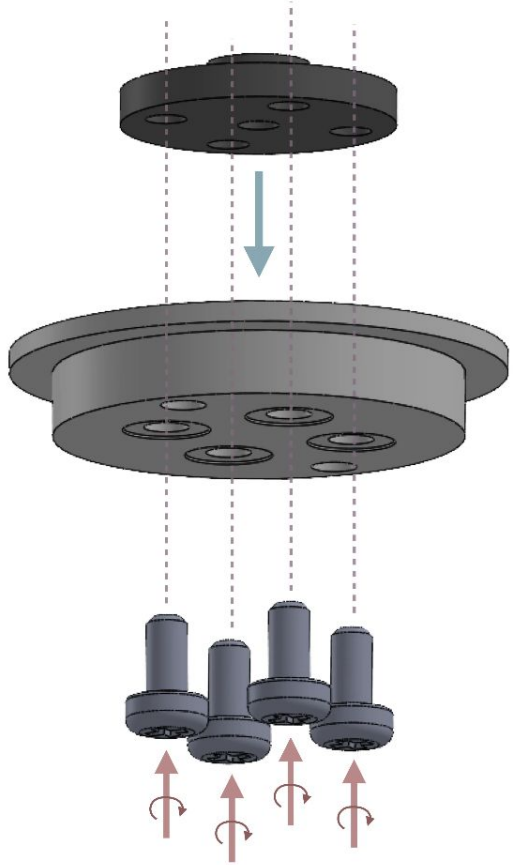


Screw in

Step B-6: Prepare Head Bearing Flange

You need:

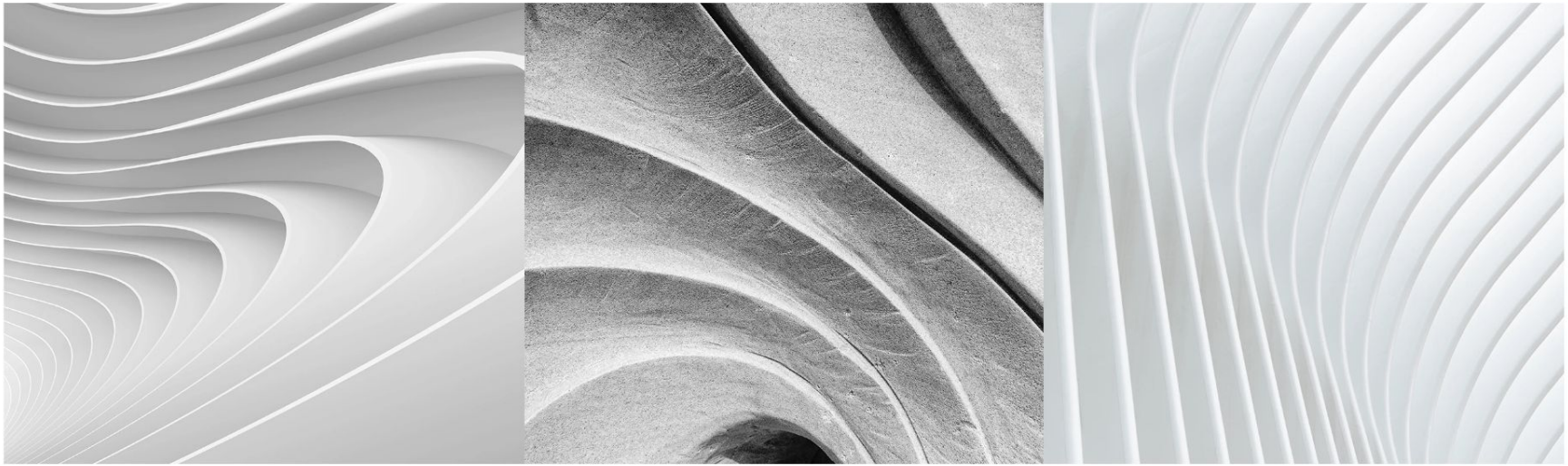
1x Head Rotation Bearing Flange
1x Servo Disk
4x M3 * 6 Screw



Screw in

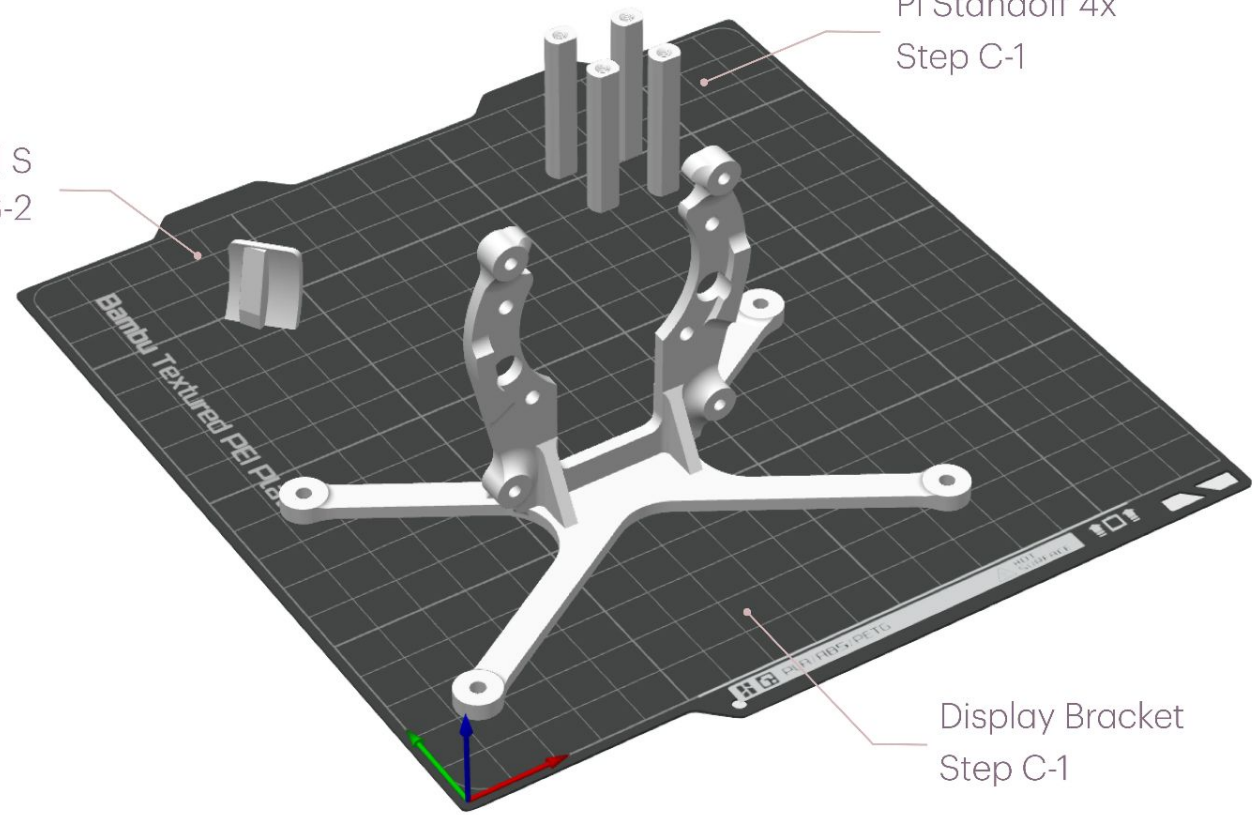
Press Fit

Part C: Head Assembly



Step C-0: 3D Print Parts

Batch #1

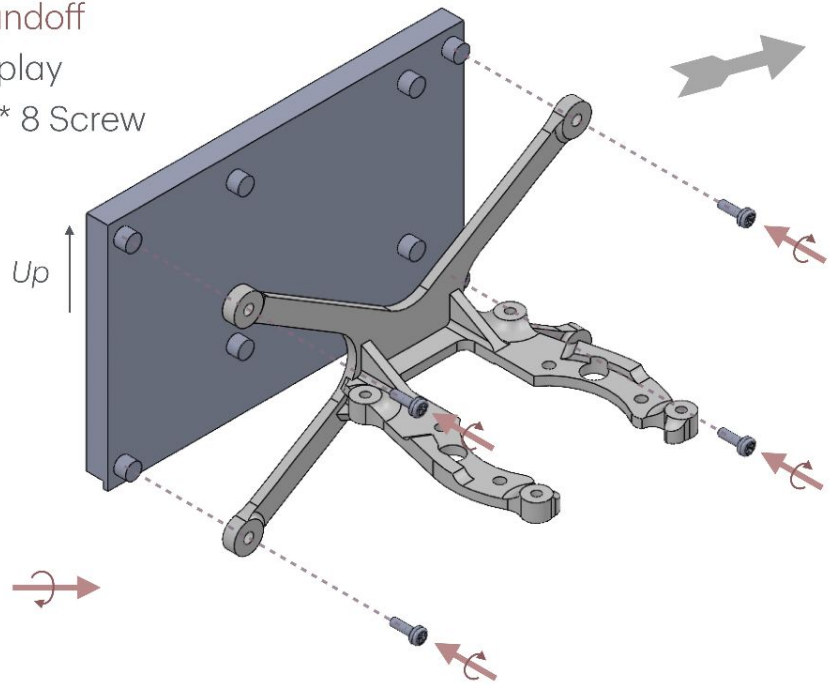


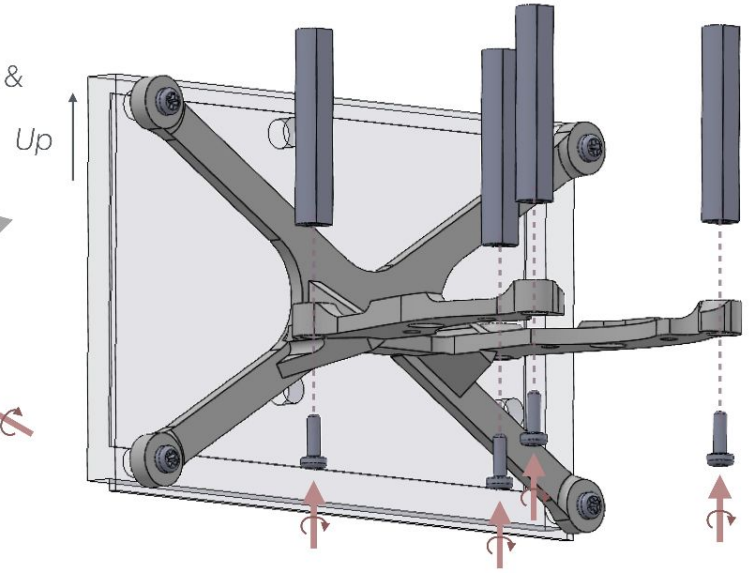
Step C-1: Attach & Build Display Bracket

You need:

- 1x Display Bracket
- 4x Pi Standoff
- 1x 5' Display
- 4x M2.5 * 8 Screw

Important: Check display orientation (up & down) before mounting the bracket



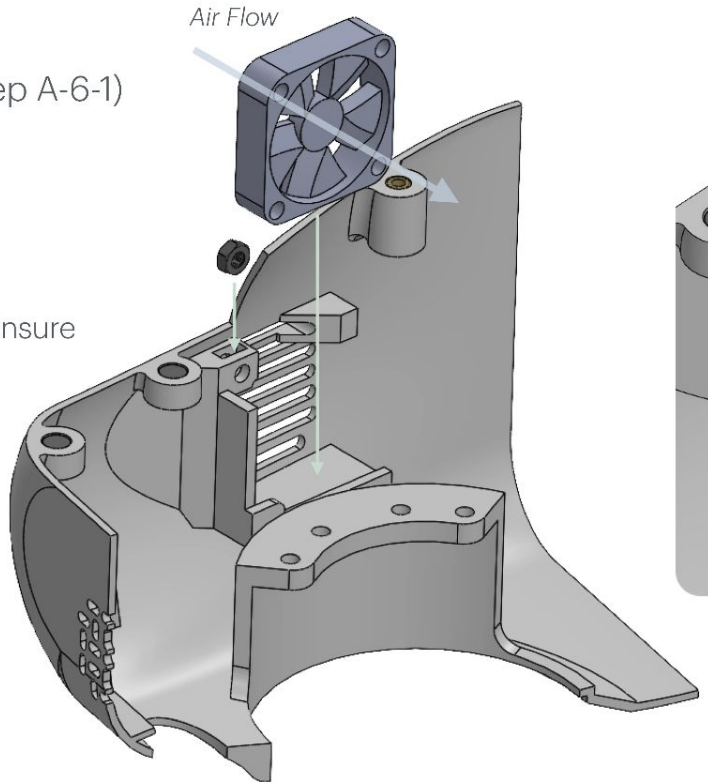


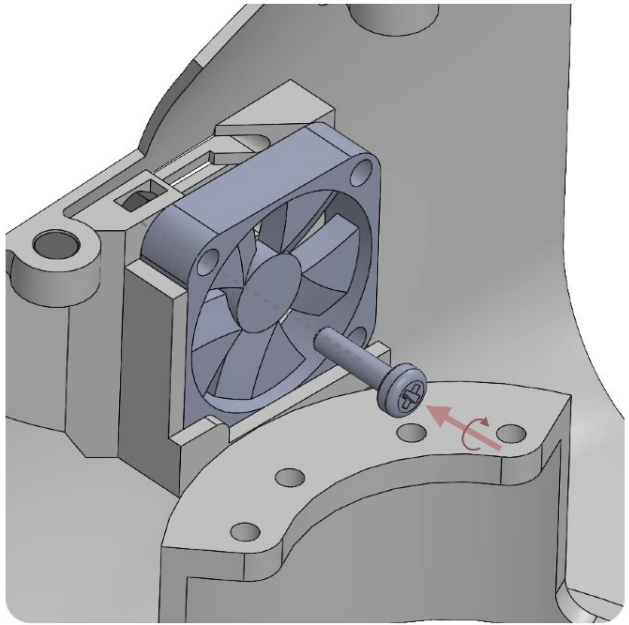
Step C-2: Attach Exhaust Fan

You need:

- 1x Head Shell L (from Step A-6-1)
- 1x Exhaust Fan
- 1x M3 Nut
- 1x M3*12 Screw

Important: Check airflow direction, ensure air is blowing into the head chamber





Screw in

Seat

Step C-3-1: Solder Switch

You need:

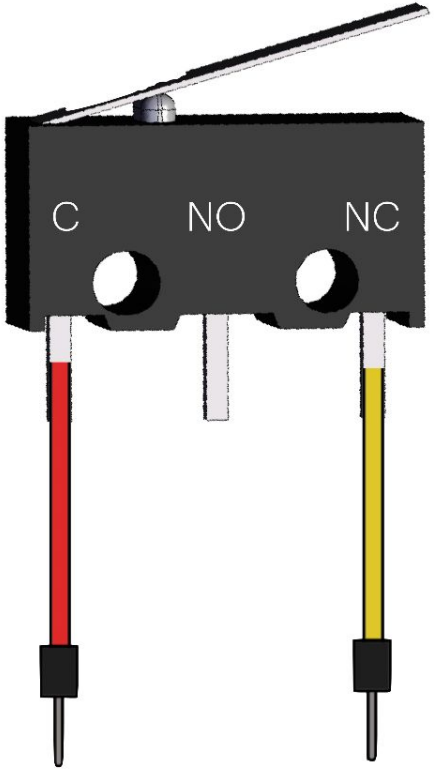
1x Micro Switch

2x Striped Jumper Wire (male)

Solder jump wires to **C** and **NC** terminals, check connectivity and apply proper insulation

Need to check!!!

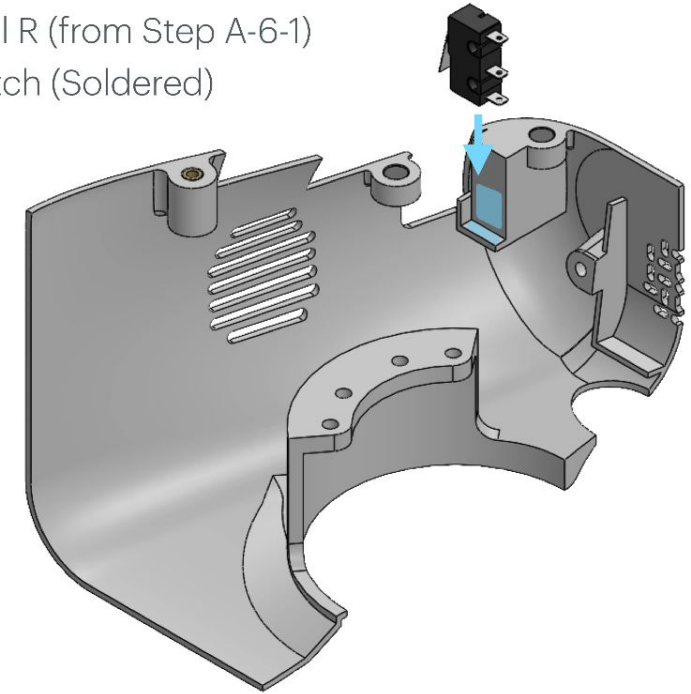
*If the switch doesn't have markings, use a multimeter (continuity mode) to find which terminal is C (COM), NO (Natural Open), and NC (Natural Close).

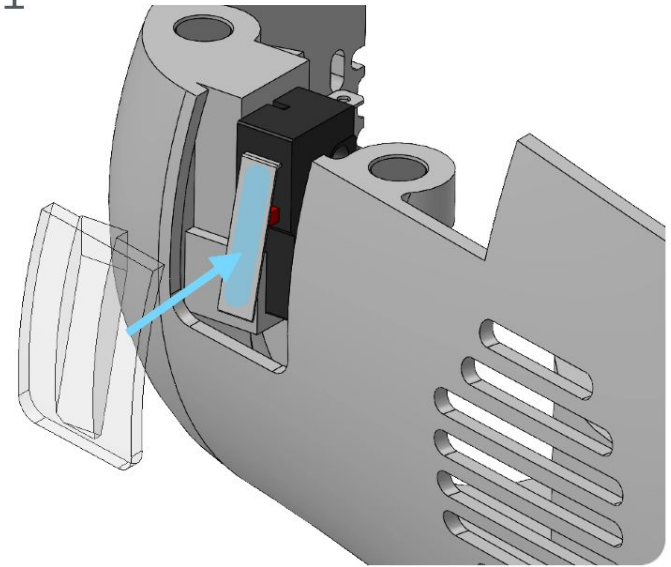


Step C-3-2: Attach Switch

You need:

1x Head Shell S
1x Head Shell R (from Step A-6-1)
1x Micro Switch (Soldered)





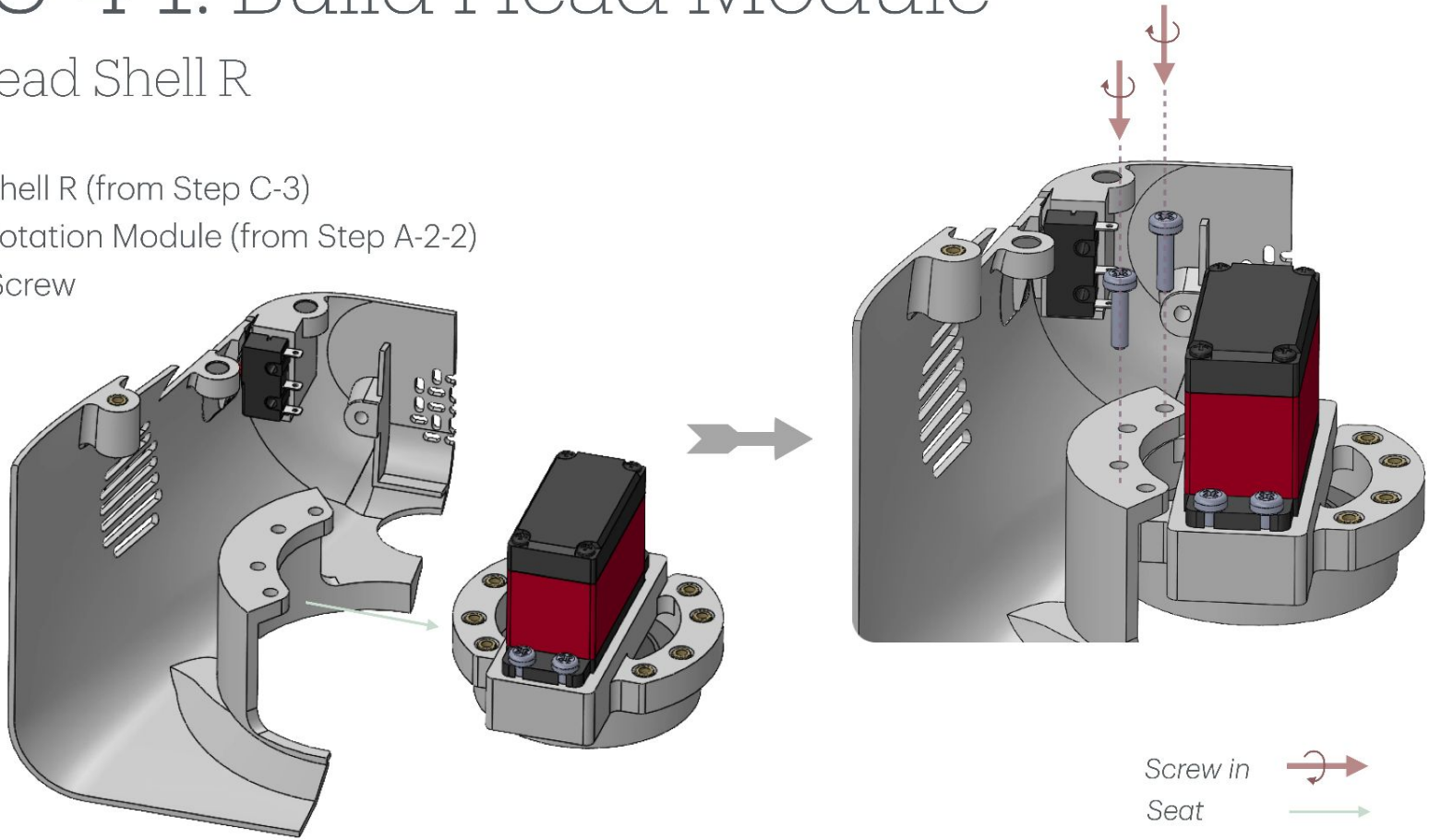
Glue

Step C-4-1: Build Head Module

Attach Head Shell R

You need:

1x Head Shell R (from Step C-3)
1x Head Rotation Module (from Step A-2-2)
2x M3*12 Screw

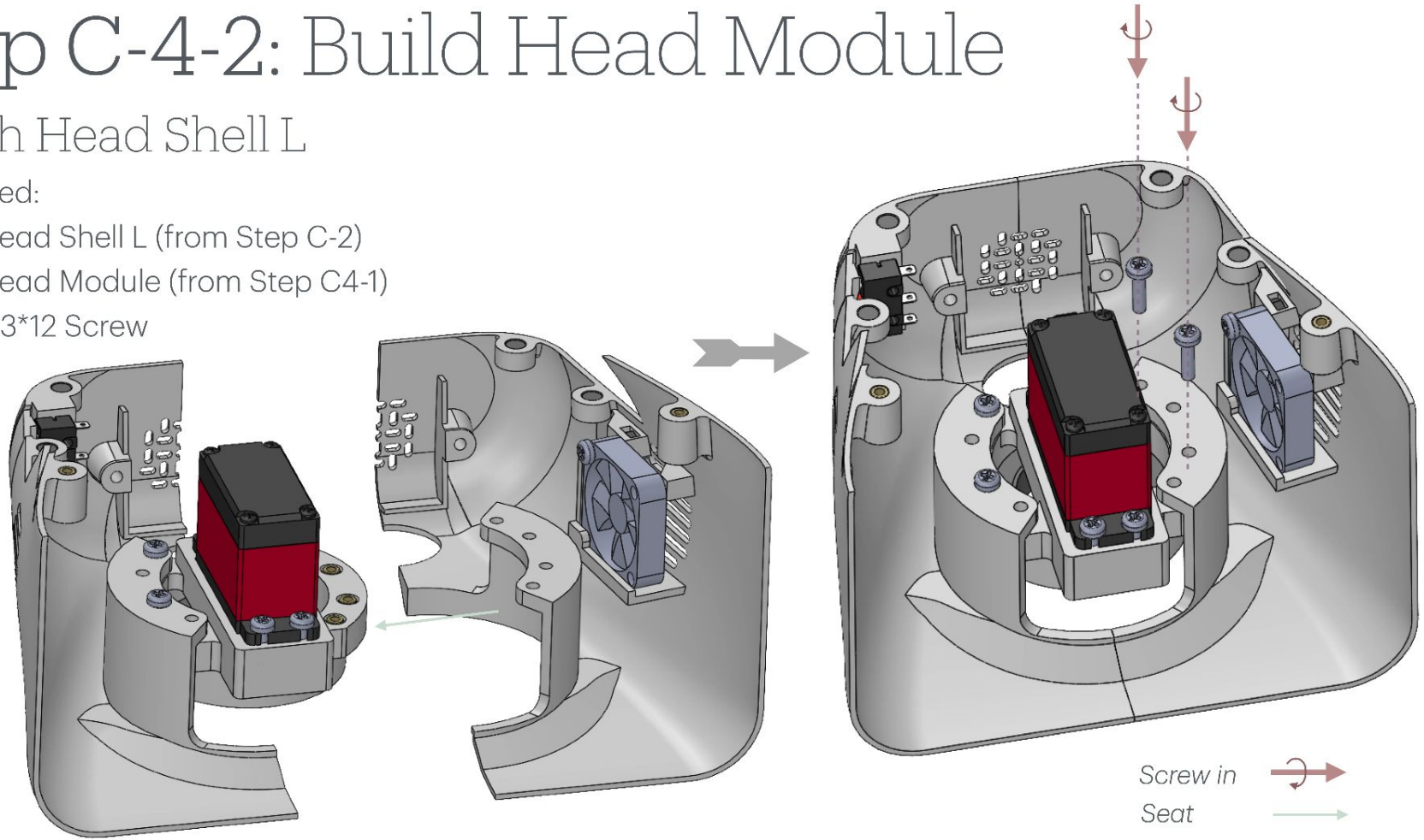


Step C-4-2: Build Head Module

Attach Head Shell L

You need:

1x Head Shell L (from Step C-2)
1x Head Module (from Step C4-1)
2x M3*12 Screw

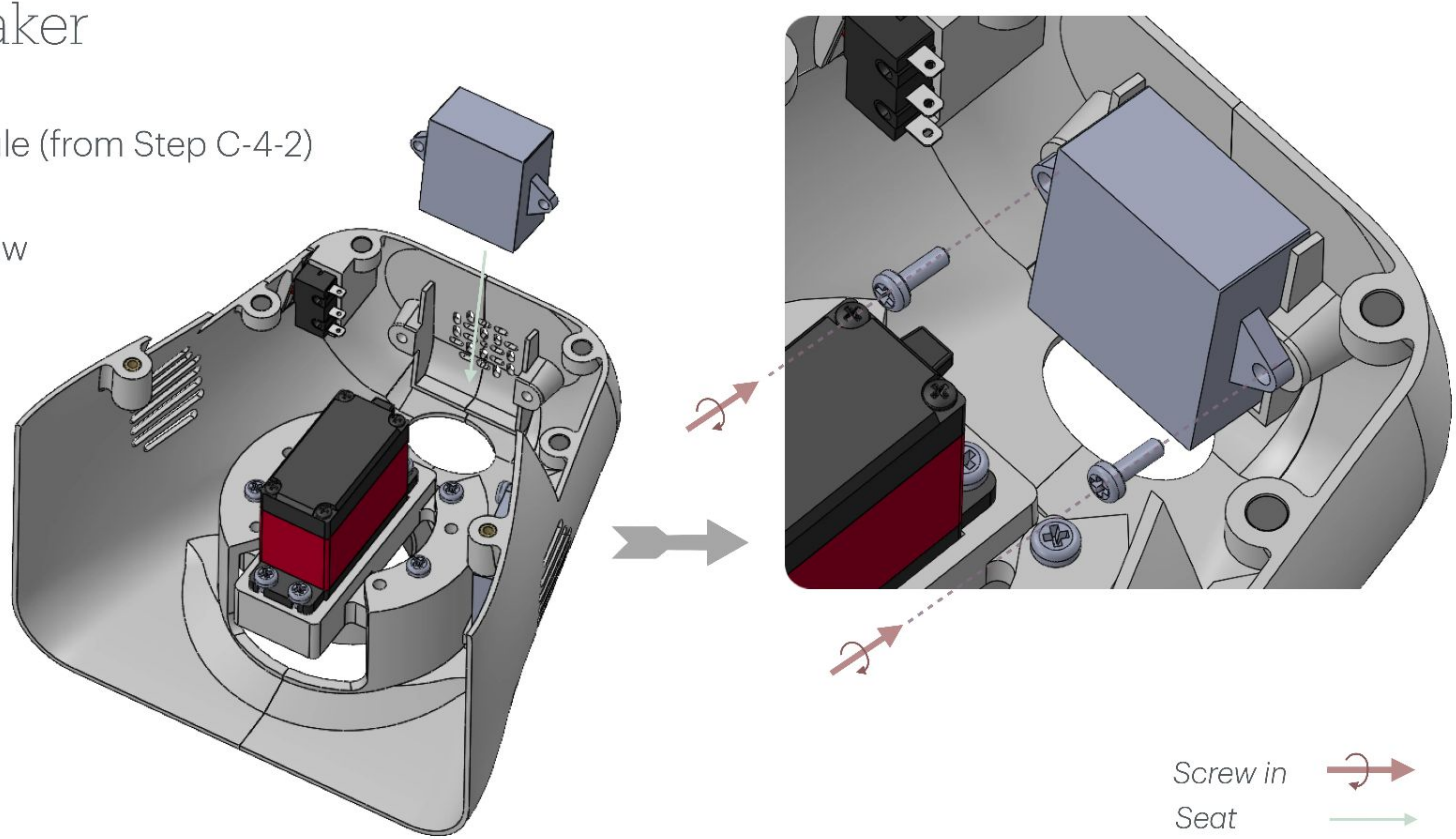


Step C-4-3: Build Head Module

Attach Speaker

You need:

1x Head Module (from Step C-4-2)
1x Speaker
2x M2.5*8 Screw

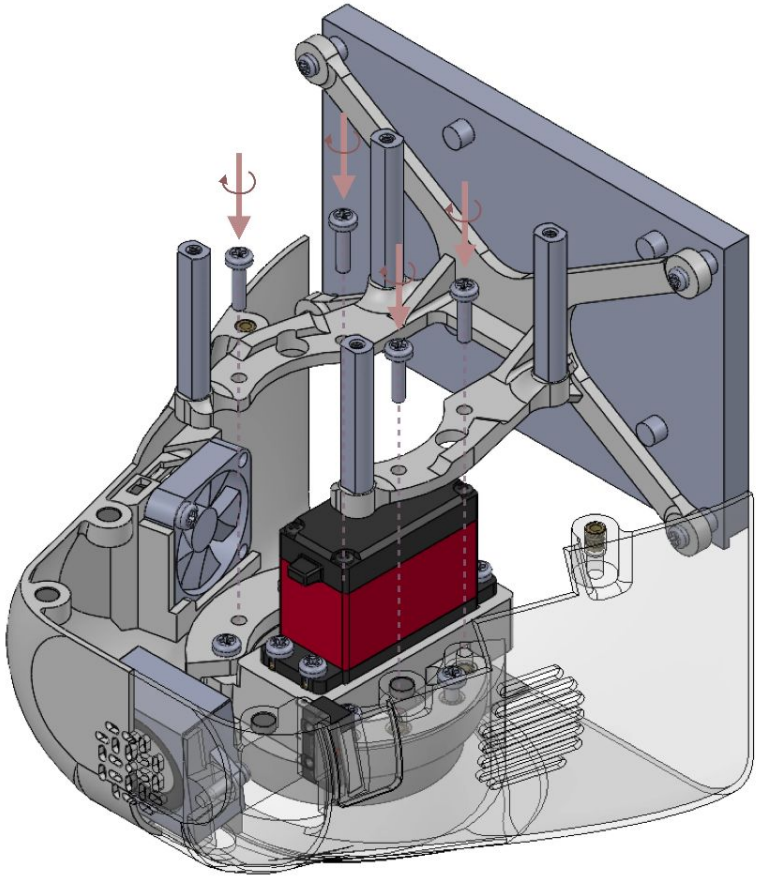


Step C-4-4: Build Head Module

Attach Display Unit

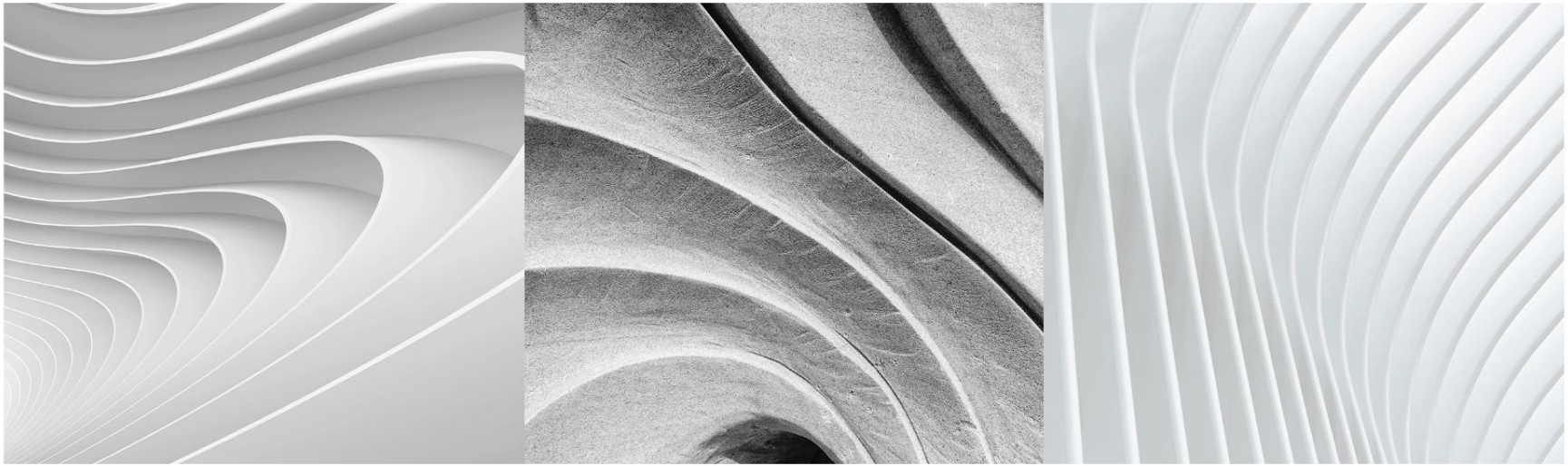
You need:

1x Head Module (from Step C-4-3)
1x Display Unit (from Step C-1)
4x M3*12 Screw



Screw in

Part D: Connecting Pi & Calibrate Servos



Step D-1: Setup Raspberry Pi and Screen

https://www.raspberrypi.com/documentation/computers/getting-started.html

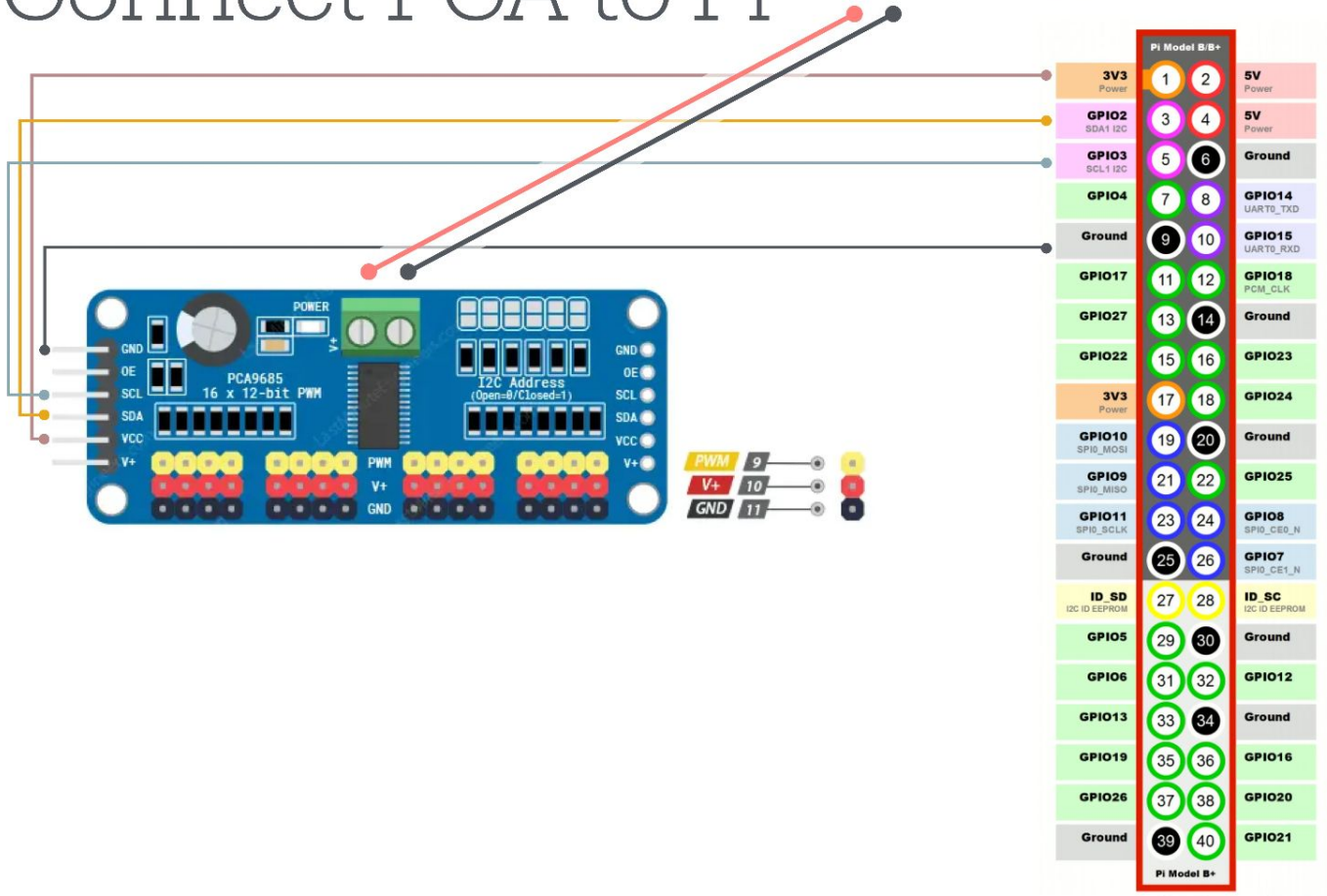
Setup the Pi (account, firmware, etc)

Connect Screen to Raspberry Pi's MIPI DSI/CSI connectors using the XXXcm Display Cable

Turn on Raspberry Pi, check basic functionality, download necessary package & code, complete basic setup

Install Library

Step D-2: Connect PCA to Pi



Step D-3: Connect Servos to PCA

Label the Pin Head of each servo

Connect the Pin heads to PCA

Record the channel each servo connects to

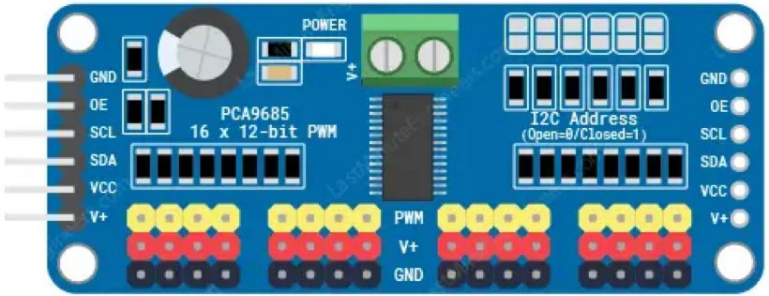
Recommended:

Channel 0 -> Body Rotation (BR)

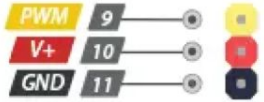
Channel 2 -> Head Nod (HN)

Channel 4 -> Head Rotation (HR)

Channel 6&7 -> Arm Rotation (AL & AR)



Channel 0 1 2 3 4 5 6 7 8 9



	BR	HN	HR	AL	AR
Channel #					

Step D-4: Run Calibration Code

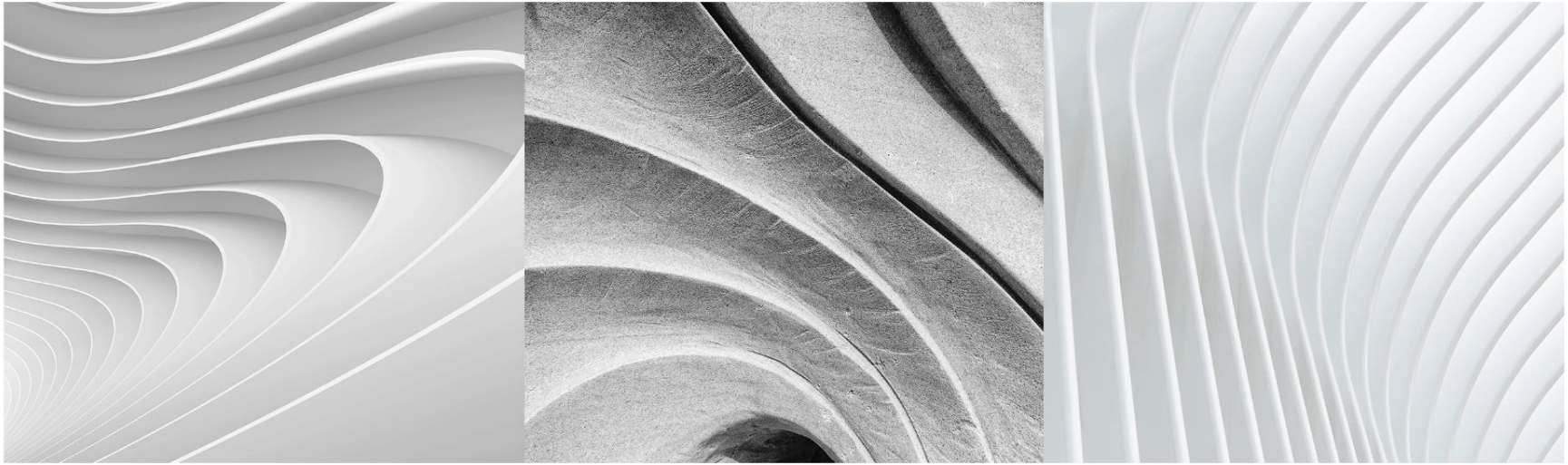
You should hear servo motor spinning

(To prevent the case that the servo is at “0” position for some reason. Let’s add a dummy script that force the servo to run back and force once)

Once Calibration finish, motors will stop

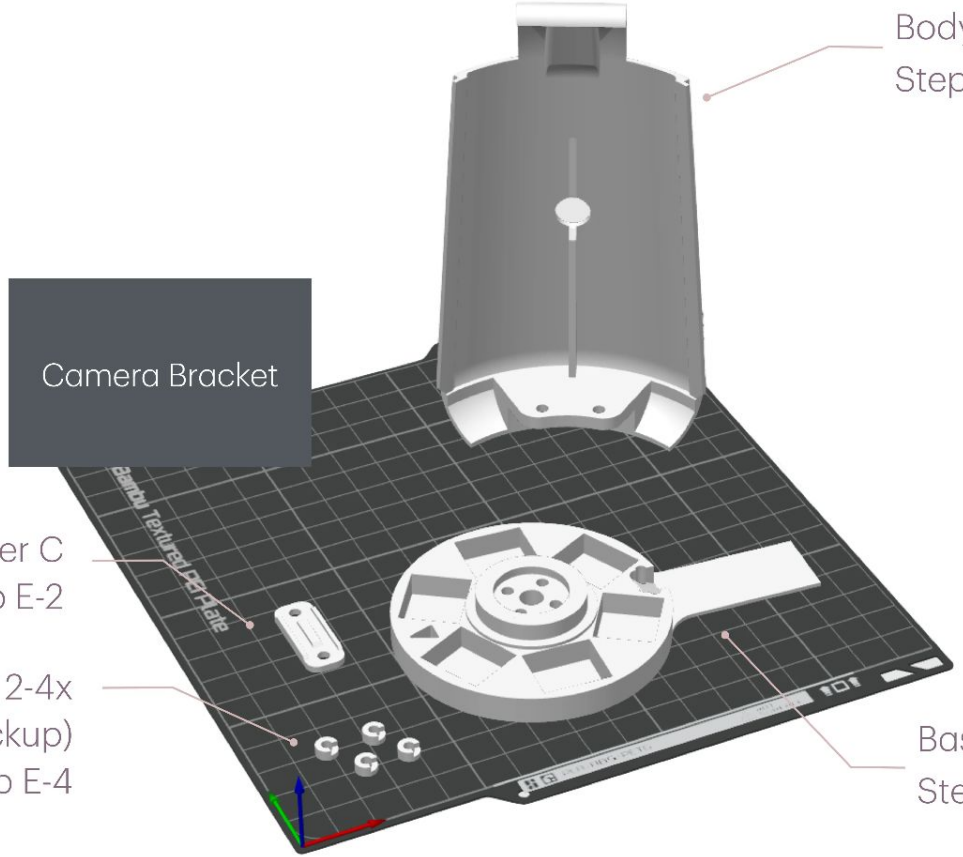
Do not unplug power (Step E would require power on)

Part E: Final Assembly



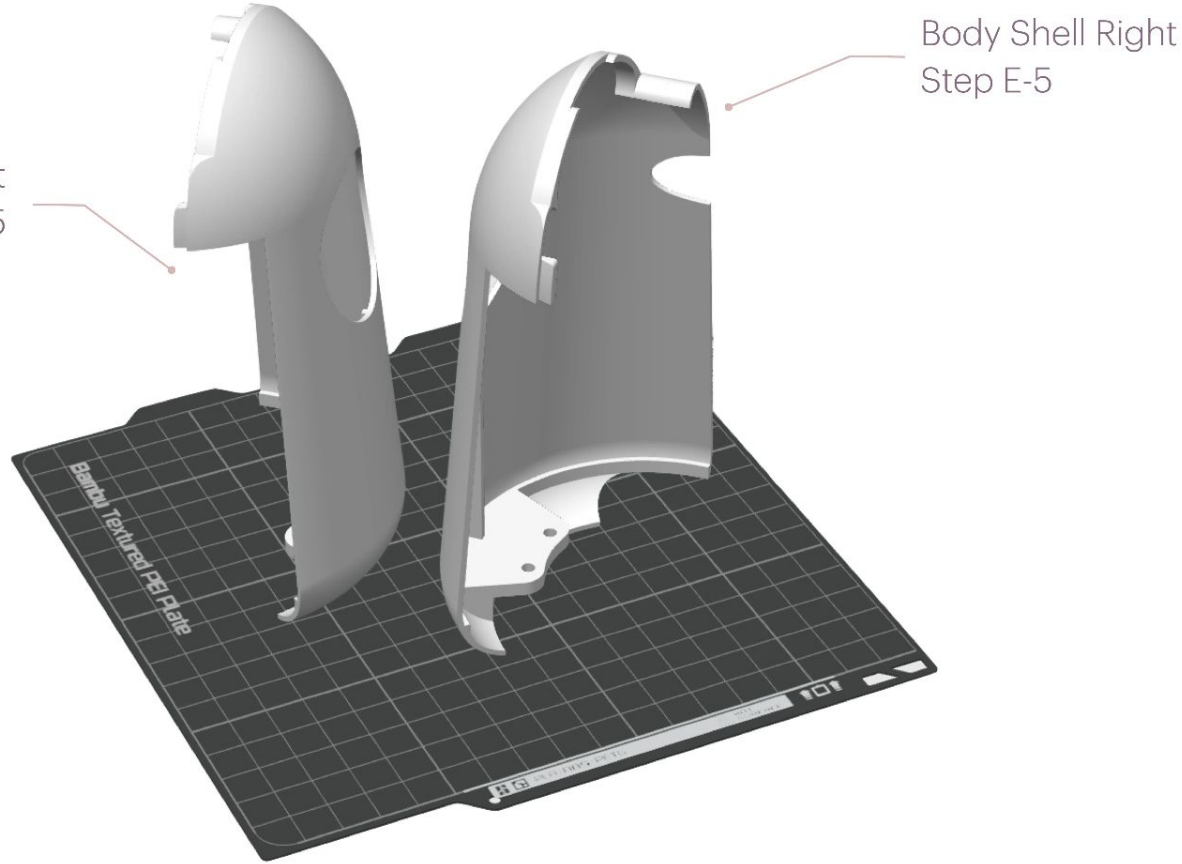
Step E-0-1: 3D Print Parts

Batch #1



Step E-0-2: 3D Print Parts

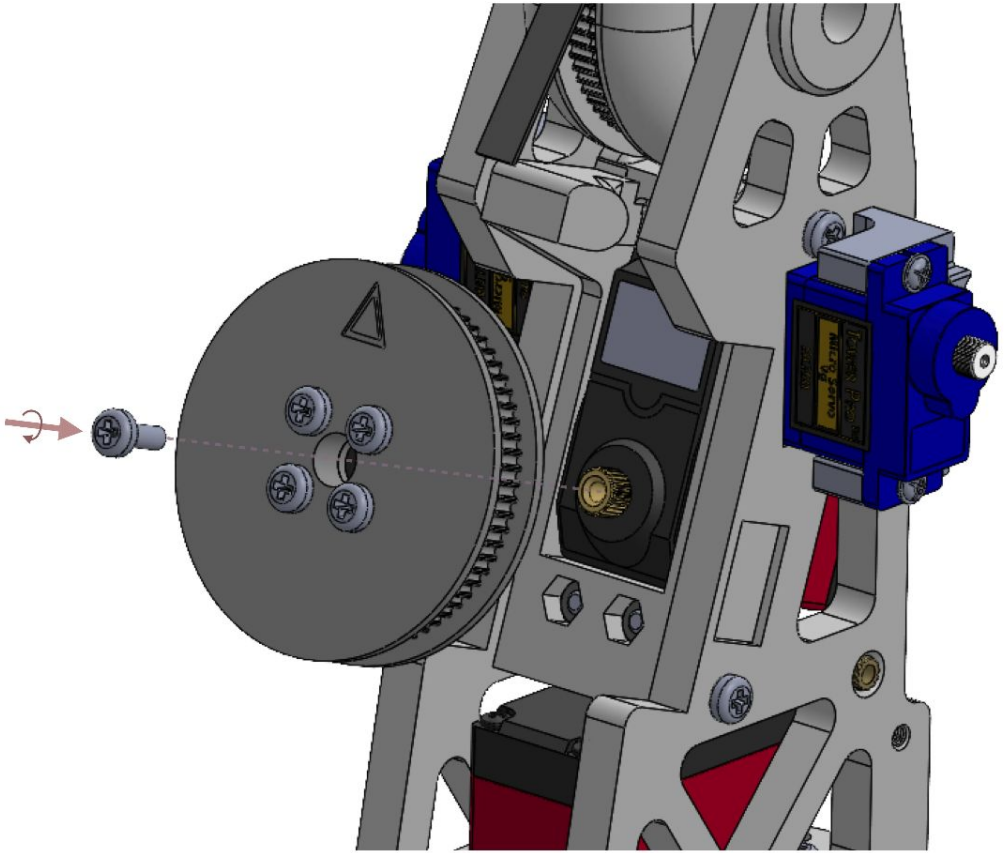
Batch #2



Step E-1: Attach Head Nod Pulley

You need:

1x Body Frame Unit (from Step B-4-2)
1x Head Nod Pulley (from Step B-5)
1x M3*8 Screw



Screw in

Step E-2-1: Tension Pulley Belt

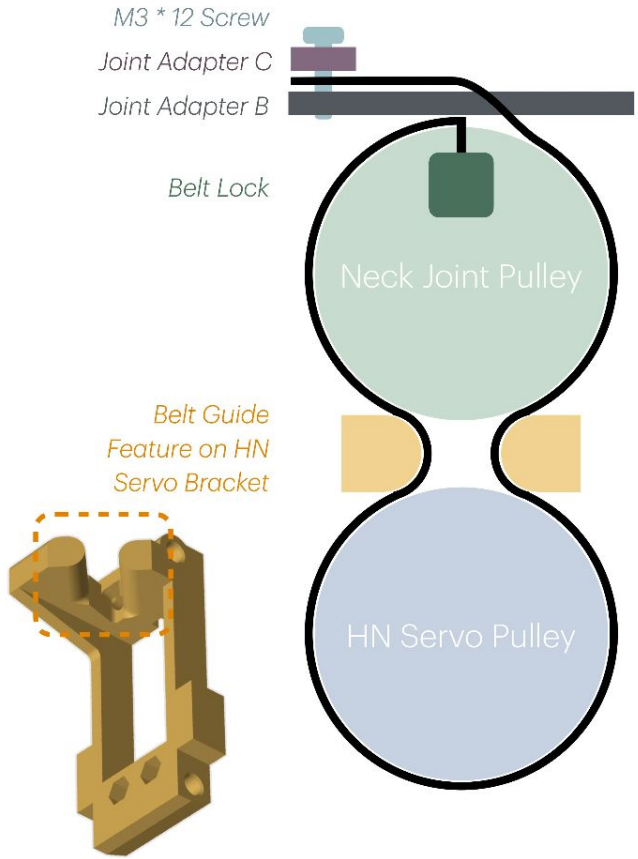
Step-1: Insert TWO M3 × 12 mm screws through the belt clamp (Neck Joint Adapter C) into Neck Joint Adapter B (from Step B-2 and B-3). Only screw them in a few turns — don't tighten yet; this leaves space for the belt to pass freely in between.

Step-2: Route the Belt

Important: Always keep tension in the belt while threading it.

Follow this exact path:

1. Start from the belt lock.
2. Run the belt along one side of the neck pulley, following the teeth.
3. Pass it through one side of the belt guide on the HN servo bracket — the smooth side of the belt should touch the guide.
4. Wrap it around the servo pulley.
   After this step, pause and check belt tension.
5. Feed the belt through the other side of the belt guide.
6. Run it along the opposite side of the neck pulley.
7. Thread it through the neck pulley plate, and between Neck Joint Adapter B and C.



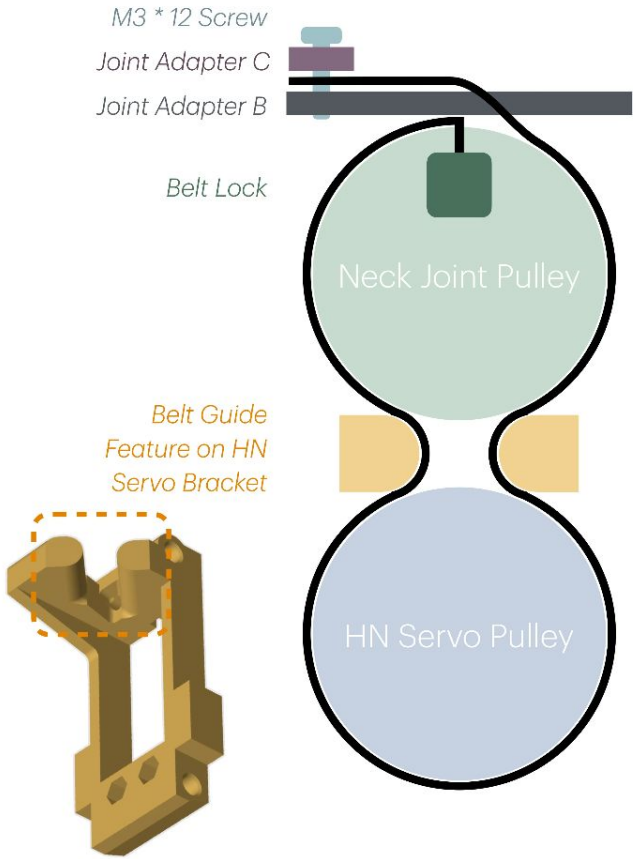
Step E-2-1: Tension Pulley Belt

Step-3: Tighten the two M3 screws from Step 1 to clamp the belt. Do not fully tighten yet — just enough to hold the belt in place.

Step-4: Verify Belt Tension

1. Disconnect 7.4 V power source
2. Manually move the neck joint to test for slippage.
3. If the belt slips, it means tension is insufficient. Reconnect 7.4 V power source then redo Step-2 to Step-3
   If no slips, congrats, Let's Move On!

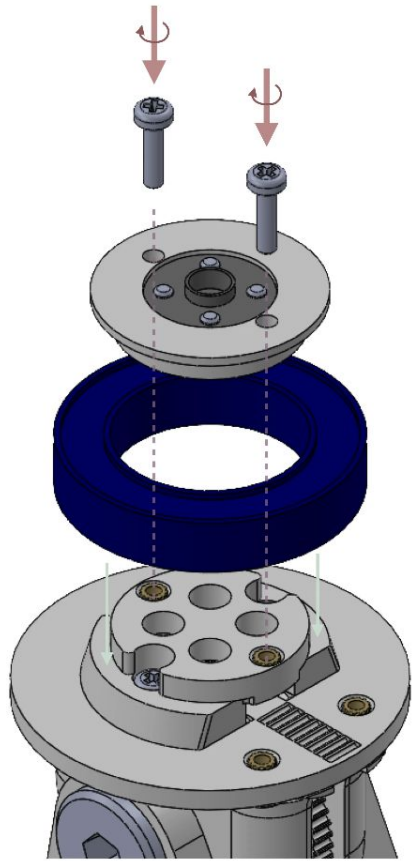
Step-5: Fully tighten the two M3 screws



Step E-3: Attach Head Rotation Bearing

You need:

1x Main Body Unit (from Step E-2)
1x Head Bearing Flange (from Step B-6)
1x 6906 Bearing
2x M3*12 Screw

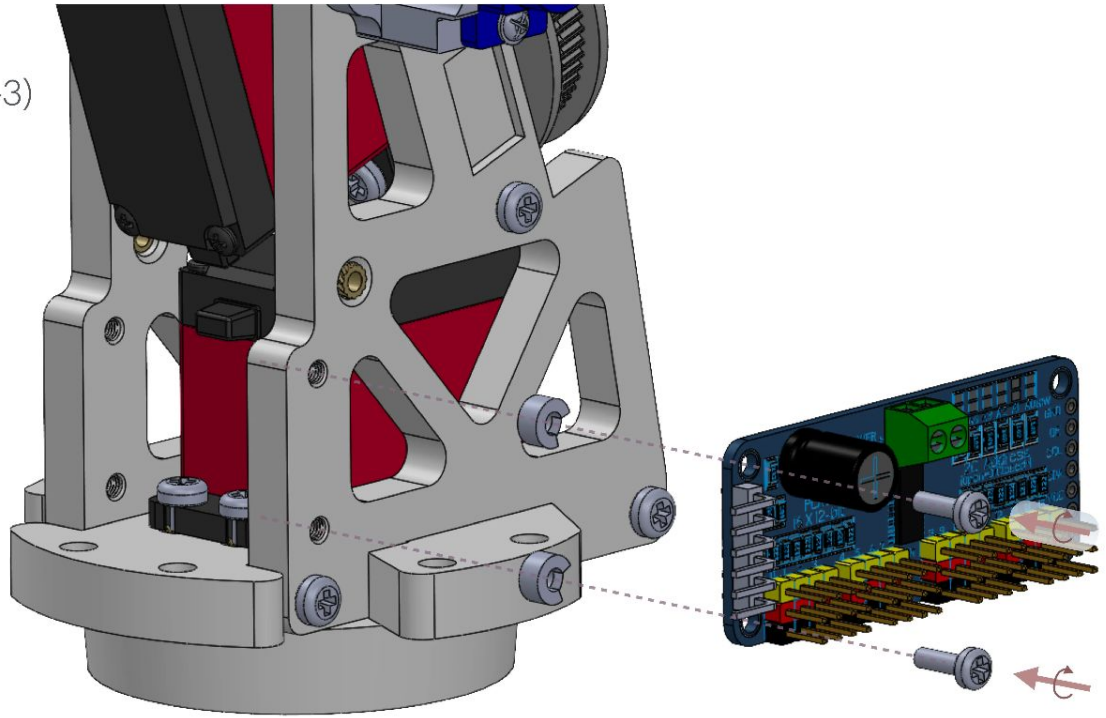


Screw in
Seat

Step E-4: Attach PCA Chip

You need:

1x Main Body Unit (from Step E-3)
2x PCA Washer
2x M2.5*8 Screw

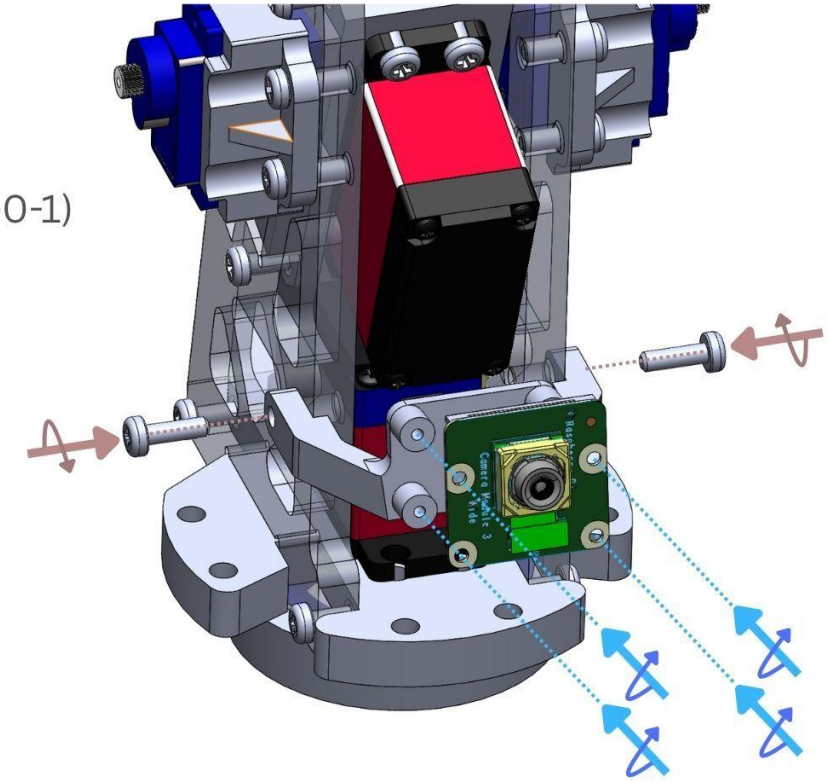


Screw in

Step E-5: Attach Camera

You need:

- 1x Camera Bracket (from Step E-0-1)
- 1x Raspberry Camera
- 2x M3*12 Screws
- 4x M2 Screws



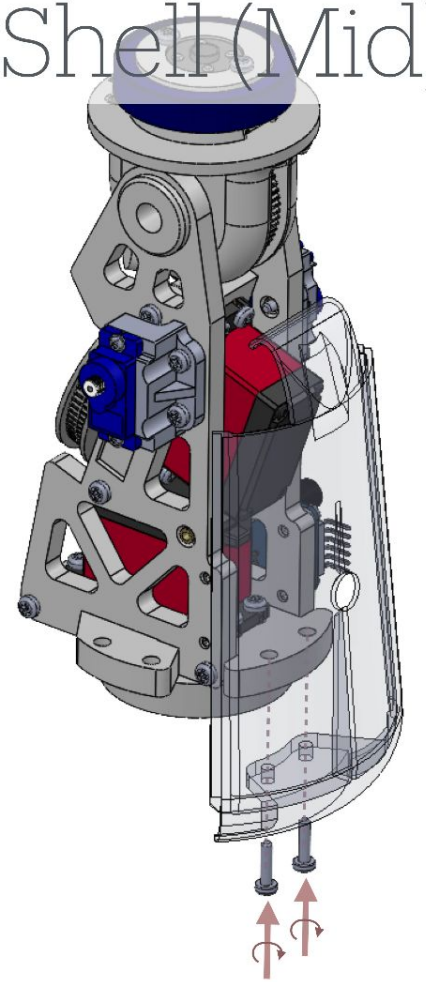
Screw in M3

Screw in M2

Step E-5-1: Attach Body Shell (Mid)

You need:

1x Main Body Unit
1x Body Shell Mid
2x M3*12 Screw

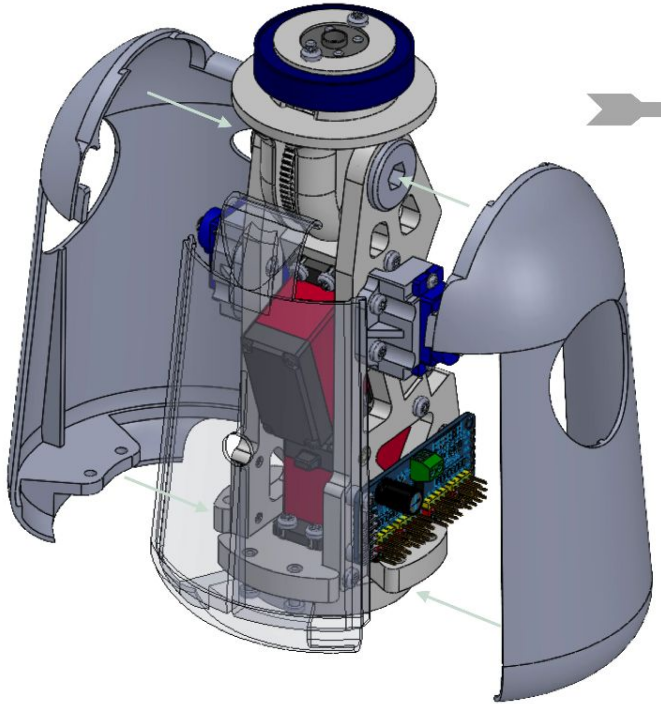


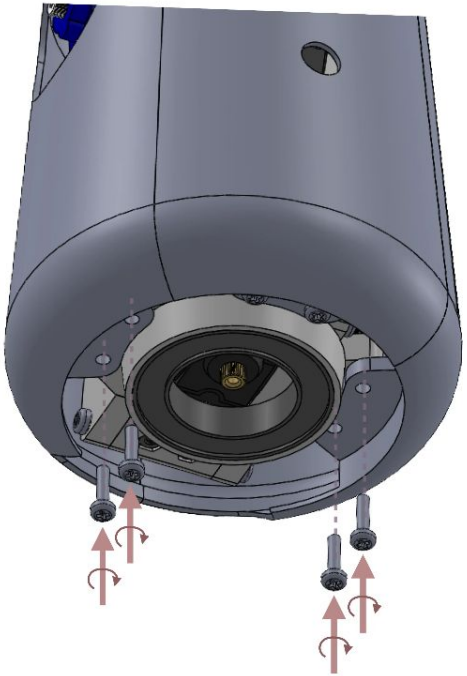
Screw in

Step E-5-2: Attach Body Shell (Left & Right)

You need:

- 1x Main Body Unit
- 1x Body Shell Left
- 1x Body Shell Right
- 4x M3*12 Screw





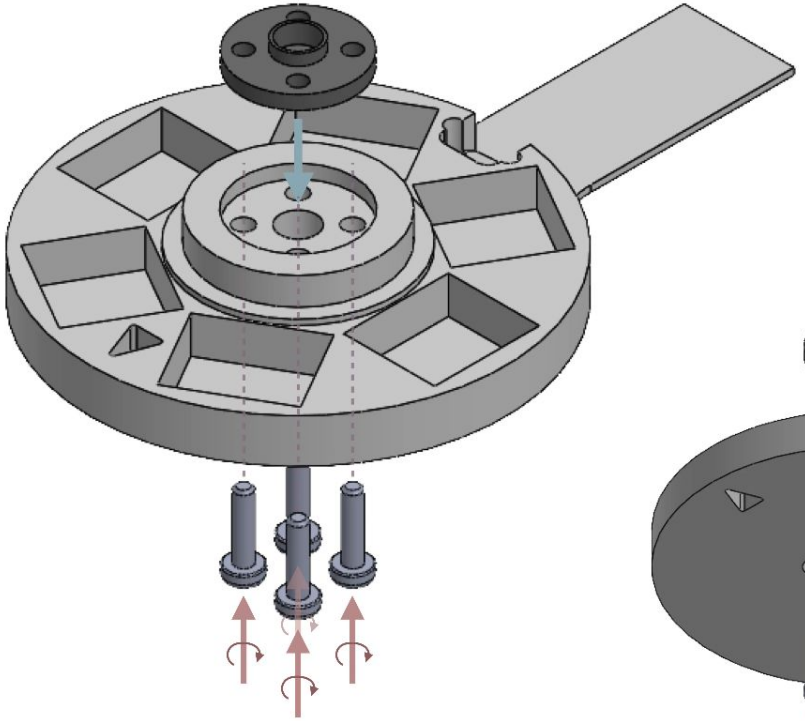
Screw in

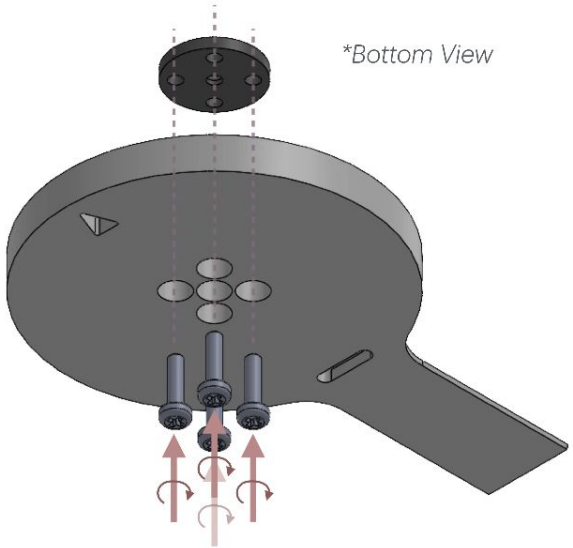
Seat

Step E-6-1: Prepare Rotation Base

You need:

1x Base Rotation Disk
1x Servo Disk
4x M3 * 6 Screw





*Bottom View

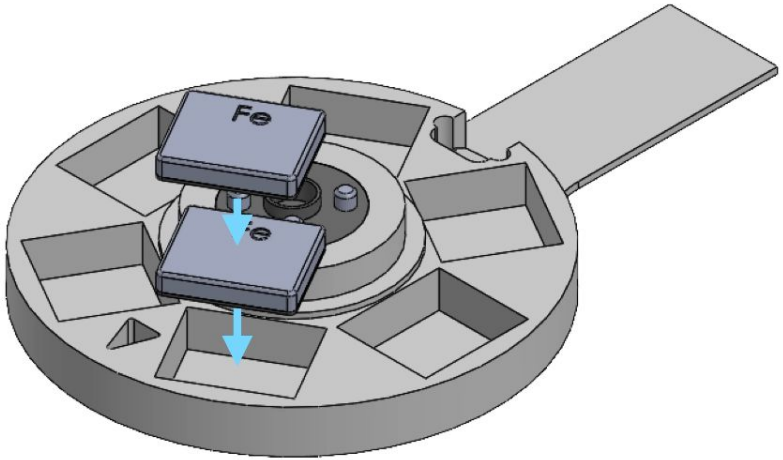
Screw in

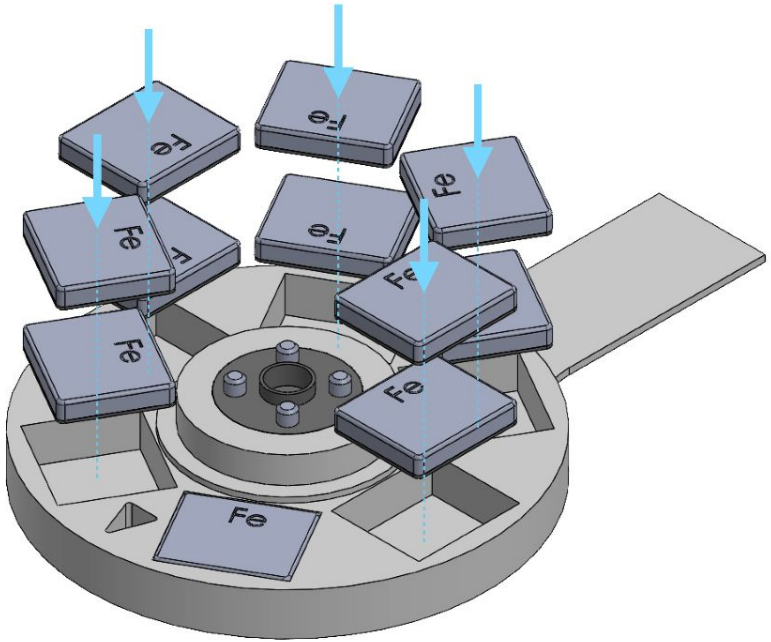
Press Fit

Step E-6-2: Prepare Rotation Base

You need:

1x Base Rotation Disk (from Step E-6-1)
1x Servo Disk
12x D5*2mm Magnets



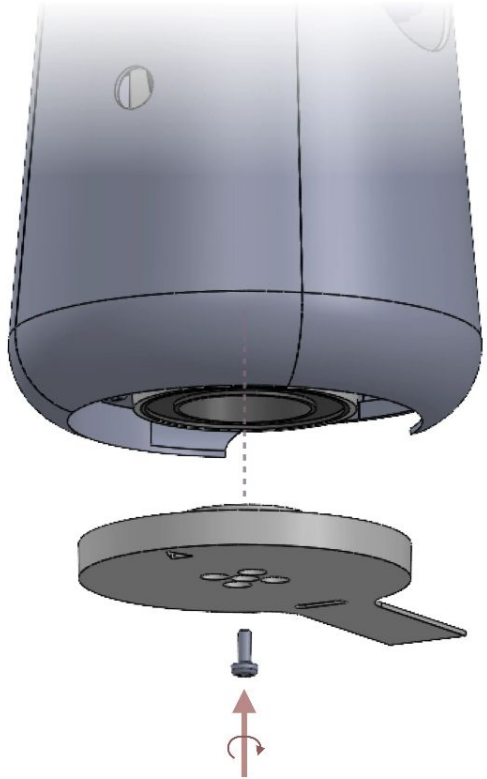


Glue →

Step E-7: Attach Rotation Base

You need:

1x Main Body Unit
1x Base Rotation Disk (from Step E-6)
1x M3*8 Screw

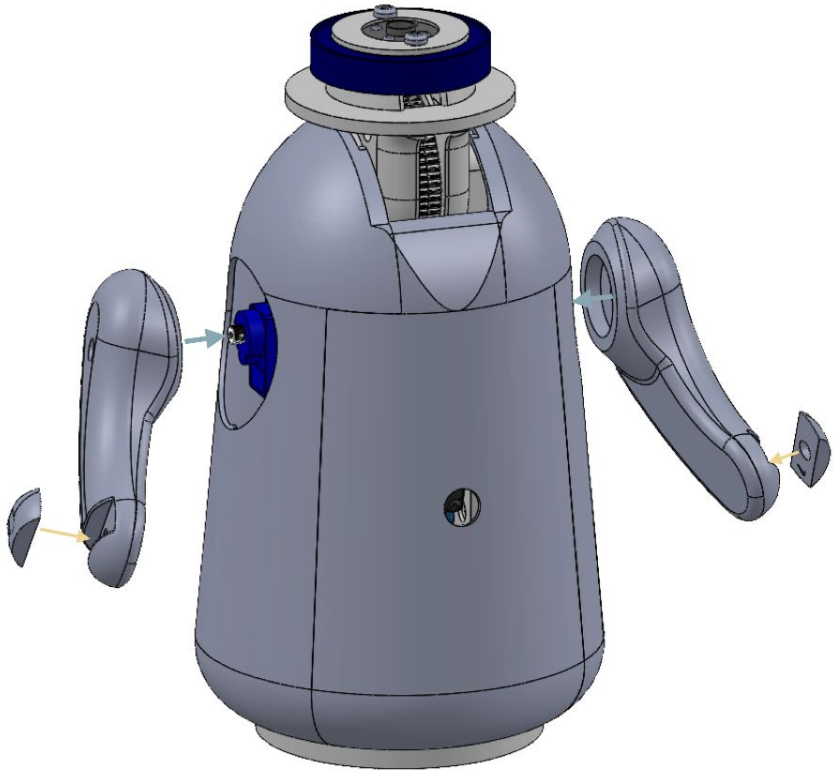


Screw in

Step E-8: Attach Arms

You need:

- 1x Main Body Unit
- 1x Left Arm (from Step A-6)
- 1x Left Arm Cover (from Step A-6)
- 1x Right Arm (from Step A-6)
- 1x Right Arm Cover (from Step A-6)



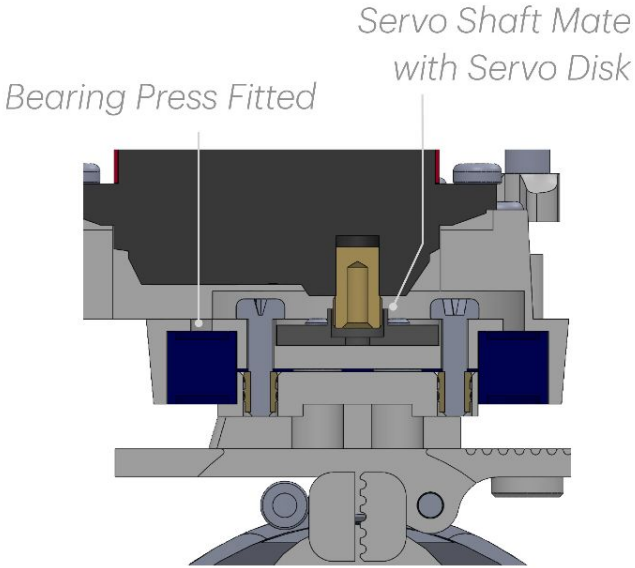
Press Fit →
Magnetic →

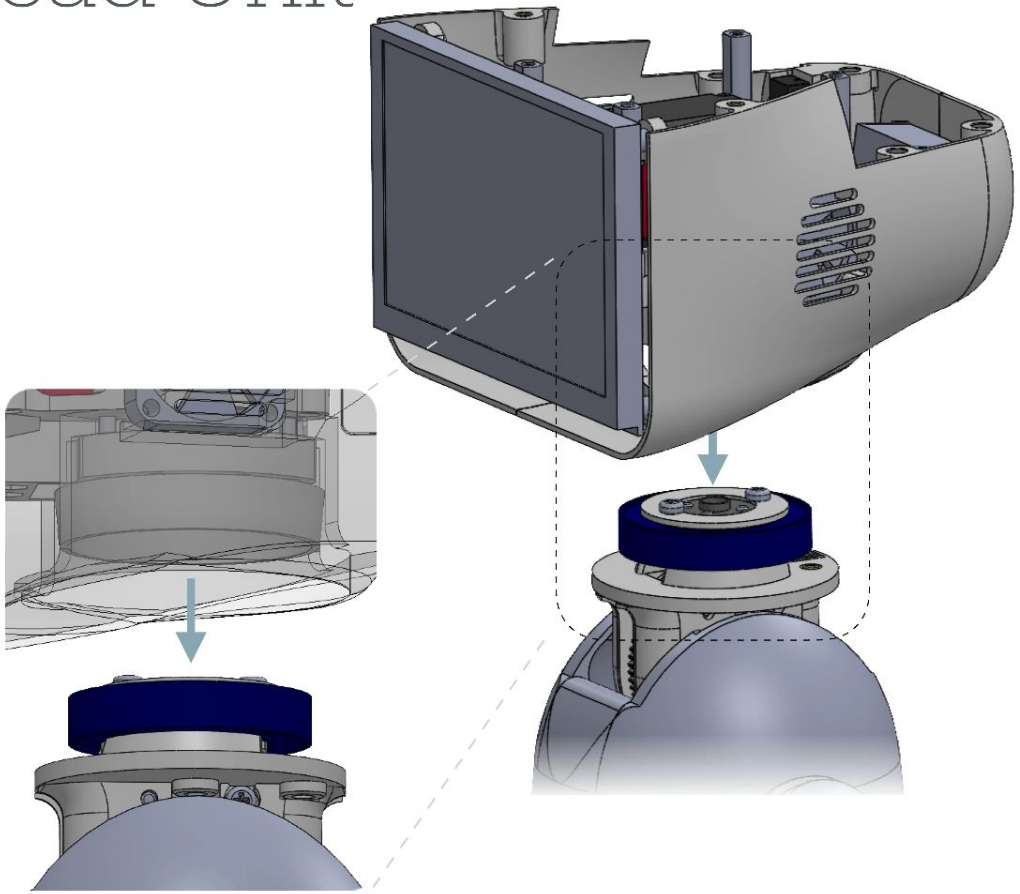
Step E-9: Attach Head Unit

You need:

1x Main Body Unit

1x Head Module (from Step C-4-4)

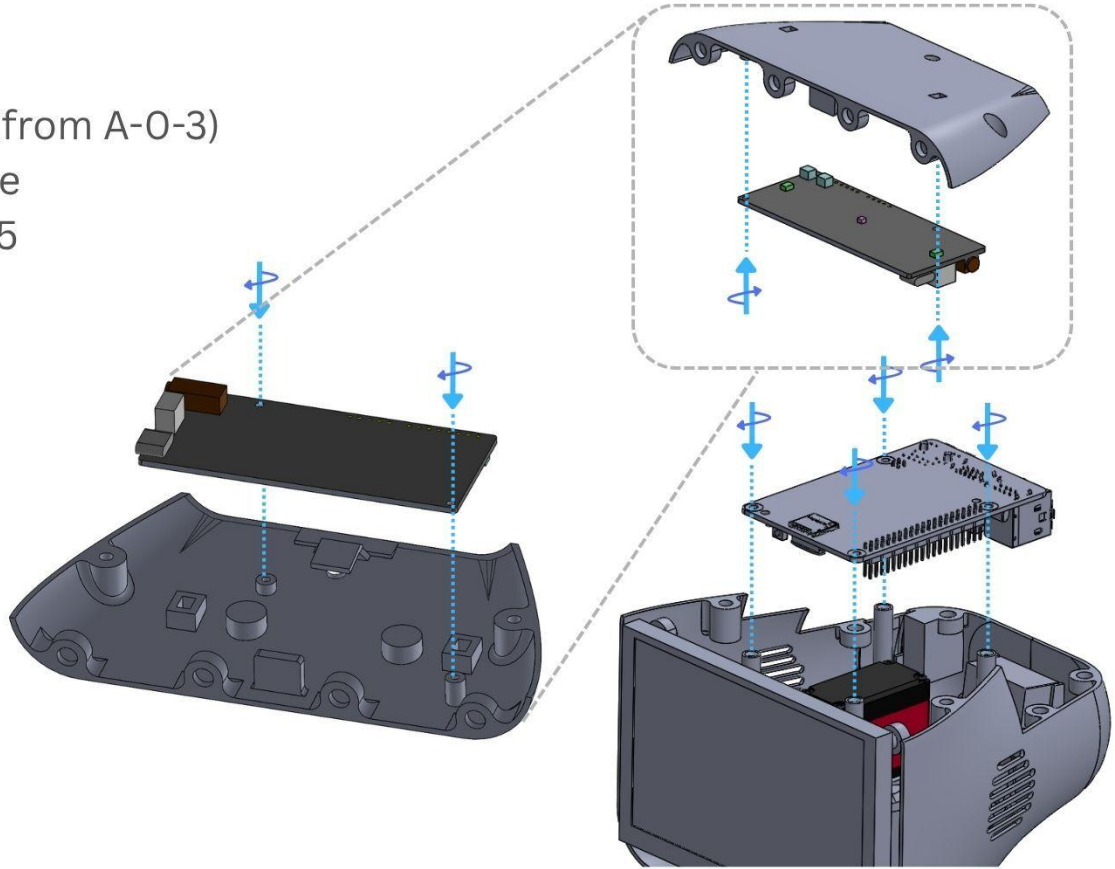




Step E-10: Attach Pi and Respeaker to Head

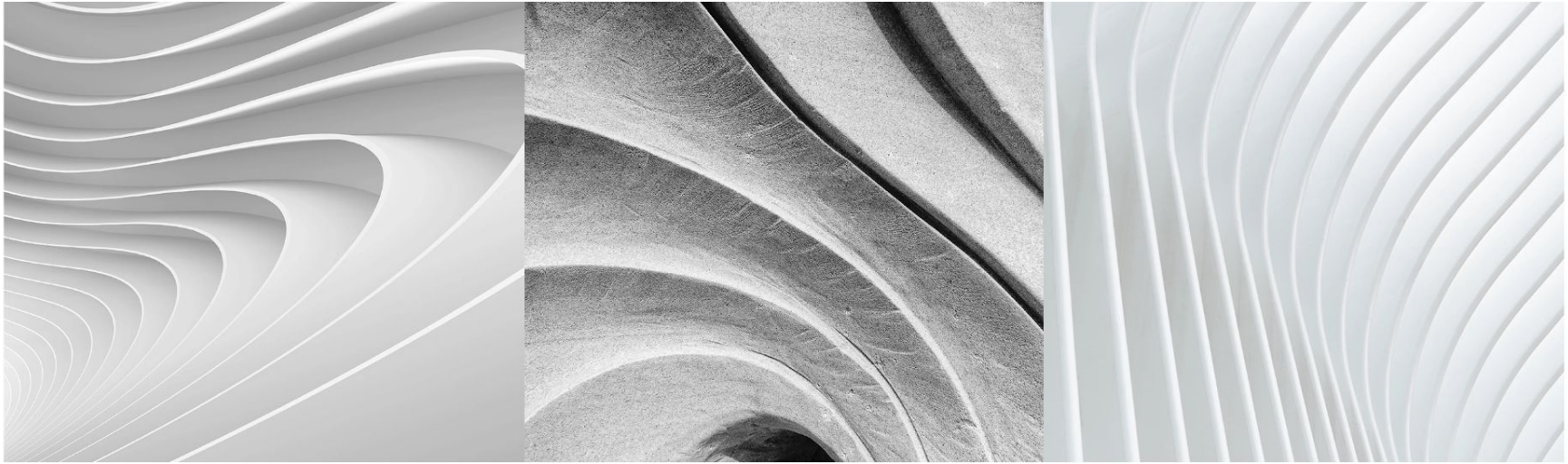
You need:

- 1x Head Shell F (from A-0-3)
- 1x Respeaker Lite
- 1x Raspberry Pi 5
- 6x M2 Screws

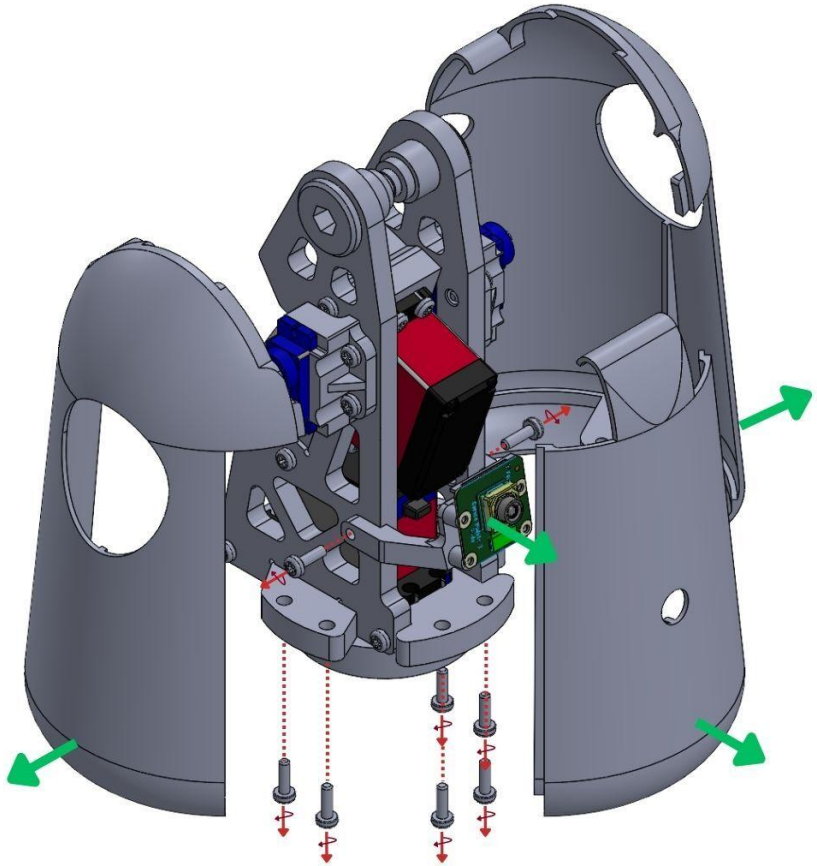


Screw in M2

Step F: Modular Extensions



Step G-1: Attach Radar Sensor



Unscrew

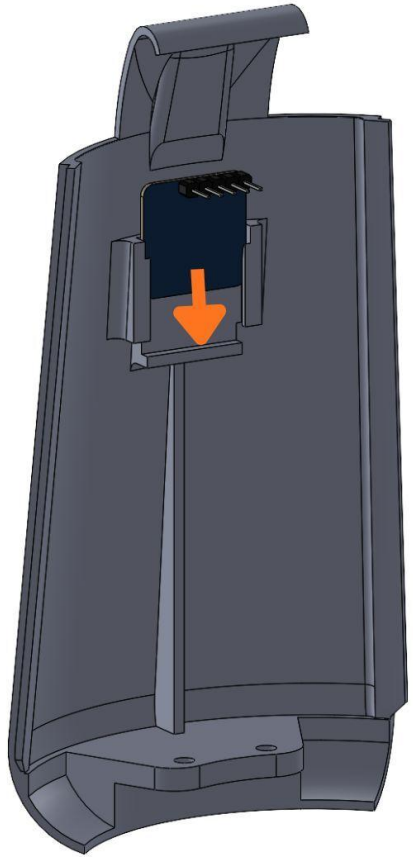
Take off

Step G-2: Attach Radar Sensor

You need:

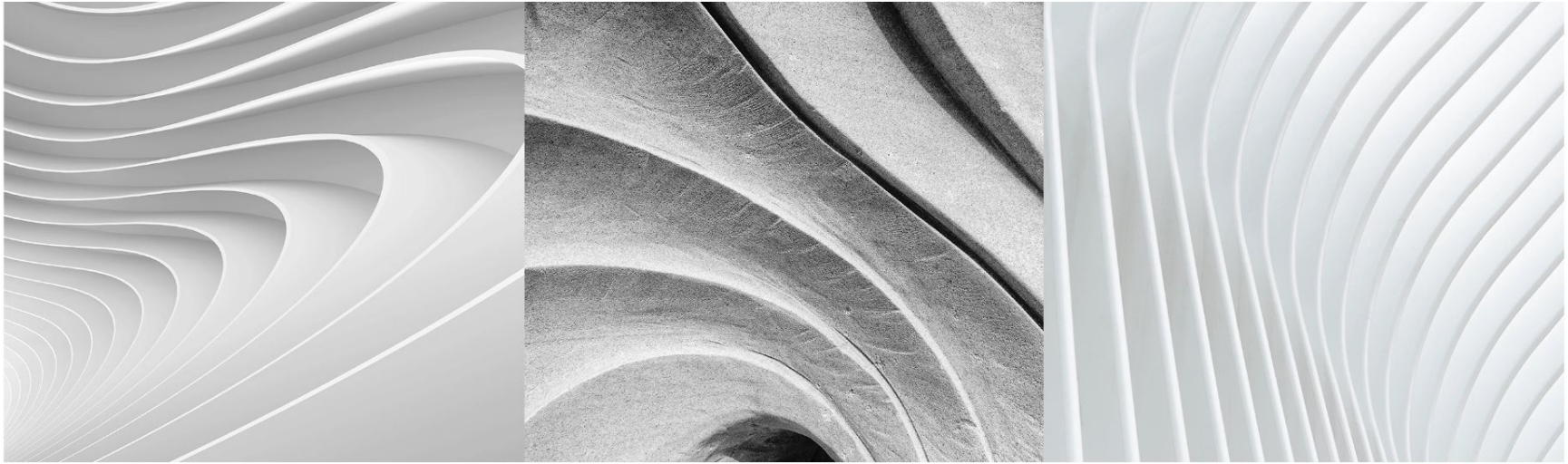
1x Body Shell Front (radar)

1x HMMD mmWave Sensor



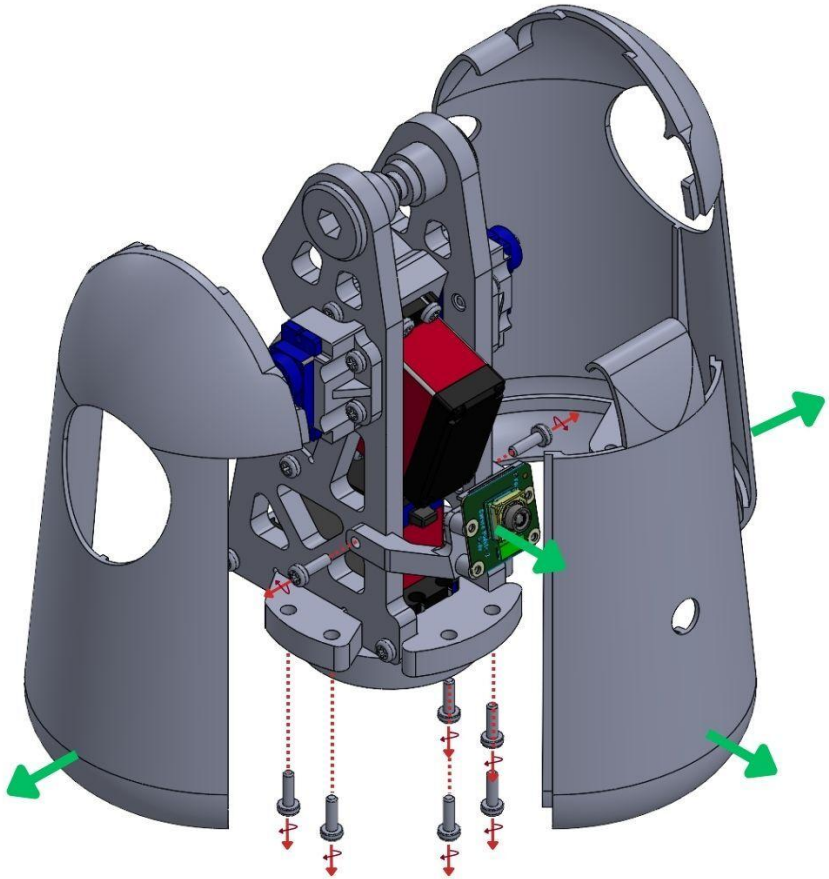
Slide fit

Step G-3: Reassemble M



Step H-1: Attach Touch Sensor



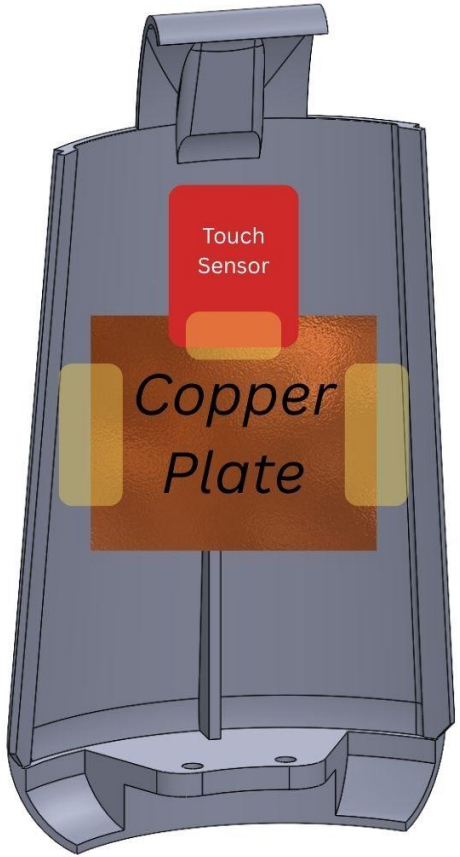


Step H-2: Attach Touch Sensor

You need:

1x Body Shell Front (no camera hole)

1x Touch Sensor



Apply Adhesive

Step H-3: Reassemble M

